PRIVATE PRINT EXPORT - 2026-08-12

An Identifiability-First Feasibility Protocol for a Gaussian-Regularized Diósi Collapse Test with a Casimir-Boundary Control

Source: docs/research/casimir-dp-quantum-foam-study.md
Source kind: canonical_docs
Source SHA-256: eb7fceb1bcf6e199da6883310b3f4cafcad09733d5b1cd654c026b2fc615e93a

Contents

Abstract

1. Introduction: question, aims, and current result
   - 1.1 Scientific context and rationale
   - 1.2 Two aims, not one blended Casimir--collapse claim
   - 1.3 Evidence tiers and present readiness
   - 1.4 What expert review is being asked to decide

2. Exact tested dynamics
   - 2.1 Registered Gaussian-regularized Diósi model
   - 2.2 Penrose OR motivation, notation, and scope
   - 2.3 Mass-density representation is part of the hypothesis

3. Leading Synthetic Commissioning Design
   - 3.1 Design selection status
   - 3.2 The dominant scale gap
   - 3.3 Platform position and required preparation receipt
   - 3.4 Operational status of the m,d,t scaling test

4. Hypotheses separated before data
   - 4.1 Ordinary-physics null, H0
   - 4.2 Frozen Diósi hypothesis, HD
   - 4.3 Lane C: boundary-conditioned extension, HB
   - 4.4 Manifold-response hypothesis
   - 4.5 Compton-frequency non-bridge

5. Complex-coherence estimator and identifiability
   - 5.1 Boundary--branch interaction diagnostic
   - 5.2 Rejected design
   - 5.3 Parent synthetic redesign
   - 5.4 Selected bounded design
   - 5.5 Material-resolved ordinary complex-coherence null
   - 5.6 Public-data component validation
   - 5.7 Proper time, worldlines, and the ordinary phase budget
   - 5.8 Superconducting boundary control: bridge and nonbridges
   - 5.9 Integrated empirical-pilot closure
   - 5.10 Retarded-source propagation closes a missing ordinary-physics lane
   - 5.11 Why the selected power is still conditional
   - 5.12 Confirmatory analysis plan still to be frozen

6. External constraints and scientific reach

7. Leading-design forecast
   - 7.1 Companion observable and interpretation ceiling

8. Commissioning, feasibility-pilot, and confirmatory decisions
- 8.1 Subsystem commissioning first
- 8.2 Integrated feasibility-pilot admission rule
- 8.3 Confirmatory campaign
- 8.4 Decisive outcomes
- 8.5 Expert collaboration work packages

9. Observable-separation gate

10. Limitations

11. Discussion
- 11.1 What is new
- 11.2 Why retain the Casimir boundary
- 11.3 What a positive result would mean
- 11.4 What a null result would mean
- 11.5 Priority order after this design study
- 11.6 Supporting derivations and design ancestry

12. Claim boundaries

13. Conclusion

Publication declarations

References

Document type: model-specific experimental design and subsystem-commissioning protocol

Version: first-reader external-review draft, 9 August 2026

Authors and affiliations: to be supplied before external submission

Corresponding author: to be supplied before external submission

External-review status: protocol/software readiness `PASS`; subsystem commissioning may proceed; integrated feasibility pilot `NOT AUTHORIZED`; 0 of 8 same-apparatus authority packets ready

> **Scientific standing.** This article does not report measured evidence for objective collapse, a physically realized mesoscopic superposition, a Casimir-to-collapse coupling, or spacetime-manifold dynamics. It reports a model-specific synthetic design result and defines the measurements required to determine whether that result survives in hardware.

> **Registration language.** In this draft, *registered* means frozen in a versioned executable repository contract. It is not a public, timestamped preregistration. A confirmatory protocol must be publicly deposited before confirmatory data are inspected.

Abstract

We propose an identifiability-first feasibility protocol for testing one specified collapse dynamics: the nondissipative, Gaussian-regularized Diósi mass-density model for a single effective particle at $R_0 = 100$ nm. Penrose objective reduction motivates the question of finite lifetime for distinct mass distributions but does not provide the registered master equation. A nearby Casimir boundary is treated only as an independently controlled electromagnetic environment. No Casimir variable enters the registered collapse generator.

The primary observable is complex center-of-mass coherence. The Diósi law and leading point are frozen, but the crossed mass--separation--hold-time grid must be qualified during subsystem commissioning and then prospectively frozen. A covariance-whitened estimator tests whether the Diósi signature remains identifiable after measured ordinary responses are projected out. A separate four-cell cross-ratio tests boundary--superposition nonfactorization; the boundary-independent Diósi factor cancels from this diagnostic when branch density and separation histories are matched.

Synthetic design screening rejected an initial nonidentifiable apparatus and found three of 200 bounded configurations that survived the current synthetic ranking filters. The leading commissioning target is a diamond-density sphere of mass 3.0925×10^{-16} kg, with 250 nm tangential branch separation, 250 ms hold time, 10 µm boundary gap, 4 K temperature, and 10^{-15} Pa pressure. The registered effective-particle model predicts 2.908% coherence loss; the lowest transported mass-density diagnostic gives 0.435%. The selected-world forecast is 1,028 dimensionless paired allocation units and conditional power 0.927 at 1,600 units. None of these values is empirical or an acquisition-time estimate.

Only bounded subsystem commissioning is presently authorized. The integrated feasibility pilot is not: state preparation and selected-specimen mass-density identity, as-built Green response, material spectra, worldline and phase covariance, gas-collision kernel, low-statistics four-cell commissioning data, an independently measurable companion channel, and an exact external-bound recast are absent. The contribution is therefore a falsifiable design and fail-closed empirical-readiness contract, not evidence for objective collapse, Casimir-modified collapse, or manifold dynamics.

Keywords: objective collapse; Diósi model; Penrose objective reduction; matter-wave coherence; Casimir boundary; identifiability; covariance whitening; experimental design

1. Introduction: question, aims, and current result

1.1 Scientific context and rationale

Matter-wave and optomechanical proposals have progressively translated collapse models from broad foundational questions into parameter-specific experimental signatures. MAQRO derives apparatus requirements from explicit science requirements; near-field nanoparticle proposals connect a preparation sequence to predicted collapse-model reach; CANNEX and MAGIS-100 make systematic-error transfer functions central to instrument design; and ARCHIMEDES separates its final vacuum-weight measurement from the pathfinder tasks required to attempt it [24--29]. This study adopts those structural lessons while addressing a different inference problem: a small coherence contraction is not evidence for collapse when the same complex-data direction can be produced by ordinary electromagnetic, thermal, gas, gravitational, control, or readout processes.

Penrose's objective-reduction argument motivates the question of whether distinct mass distributions have a finite lifetime [4,5]. The numerical prediction here comes instead from one named Gaussian-regularized Diósi generator [1--3]. They are not the same theory object. The Casimir boundary is retained because it creates a tunable electromagnetic environment that can stress-test the ordinary-physics model; no Casimir variable enters the registered collapse generator.

The experiment asks:

> After ordinary thermal, electromagnetic, gas, vibration, readout, relativistic phase, and boundary-correlated responses have been measured, can the remaining complex-coherence data support or exclude the prespecified mass--separation--hold-time signature of one regularized Diósi model?

This is intentionally narrower than “does gravity collapse the wave function?” A sensitive null can constrain only the registered implementation and parameter region. A positive residual can be associated with that implementation only when it is identifiable against the measured nuisance span and survives the prespecified companion and replication rules. Neither outcome establishes Penrose's spacetime interpretation.

1.2 Two aims, not one blended Casimir--collapse claim

The apparatus answers two orthogonal questions:

1. **Primary aim -- named-model scaling.** Test the boundary-independent Diósi coherence contraction across mass, branch separation, and hold time after ordinary-background closure.
2. **Secondary aim -- boundary interaction.** Test whether the boundary and material superposition fail to factorize after the measured electromagnetic response is removed. Standard boundary-independent Diósi attenuation cancels from this estimator only when the compared cells have matched mass- density and branch-separation histories.

Leading commissioning target: diamond-density sphere; radius 276.302 nm; mass 3.0925×10^{-16} kg; 250 nm tangential separation; 250 ms hold; 10 μm boundary gap; 4 K; 10^{-15} Pa. These are design targets, not achieved apparatus specifications.

Test or gate	Experimental contrast	Observable or frozen expectation	Defensible inference	Present authority
state feasibility	compact versus verified separated packet	packet centers, covariance, overlap, momentum difference, fidelity, and recombination	failure is an apparatus no-go, not a collapse-model result	no preparation at the registered mass, separation, and hold time has been demonstrated
Penrose missing-theory candidate	formal branch objects under one proposed correspondence versus ordinary unitary evolution	Stage-0.1 diagnostic correspondence residuals and weak-field recovery only; no physical rate or prediction vector	a powered test could constrain only a separately completed and frozen candidate	the synthetic correspondence benchmark passes, but 0 of 5 physical reference packets are ready; the parent remains blocked at the scientific branch-correspondence rule
primary Diósi test, H_{D}	prospectively frozen mass, separation, and hold-time cells	$C = Ve^{i\phi}$; 2.908% named-model loss and 0.435% transported diagnostic benchmark at the leading point	a sensitive null constrains the registered region; a replicated matching contraction is model-consistent phenomenology	complete m, d, t allocation, complex coherence, nuisance responses, and covariance remain unmeasured
boundary interaction	active/reference boundary × separated/compact branches	four-cell ratio R_4; standard boundary-independent Diósi loss cancels for matched mass-density and separation histories	corrected $R_4 \neq 1$ establishes a boundary--superposition anomaly, not Casimir-induced collapse	measured Green/FDT response and the four same-apparatus coherence cells are absent
ordinary electromagnetic null, H_0	gap, material, polarization, temperature, sham, and normal/superconducting state	signed phase and contraction from measured material response and a finite-geometry Green tensor	agreement validates the boundary calibration chain; disagreement repairs H_0 first	current values are software fixtures; specimen spectra and the as-built Maxwell solution are absent
proper-time and control phase	three-dimensional branch reversal, echo, path swap, local-mass, and rotation controls	signed unitary phase $\Delta\phi_{\mathrm{prop}} = -(mc^2/\hbar)\Delta\tau$	reversal or echo cancellation identifies ordinary phase contamination	the 0.018007 rad synthetic budget assumes unmeasured 10^{-4} echo rejection
same-generator companion	identical m, R_0 parameter point	predicted center-of-mass heating $\dot{E} = 3.07 \times 10^{-40}$ W	an independent matching channel could strengthen generator identification	no instrument-level detector receipt is supplied; the strongest positive interpretation is unavailable
external and pilot gate	variant-matched external bound plus eight empirical packets	preparation ∧ response ∧ covariance ∧ identifiability ∧ power ∧ custody	every factor must pass before confirmatory planning	scalar external screen only; 0 of 8 authority packets are ready

Figure 1 shows the proposed geometry. Figure 2 shows the acquisition and custody sequence.

Leading synthetic commissioning design

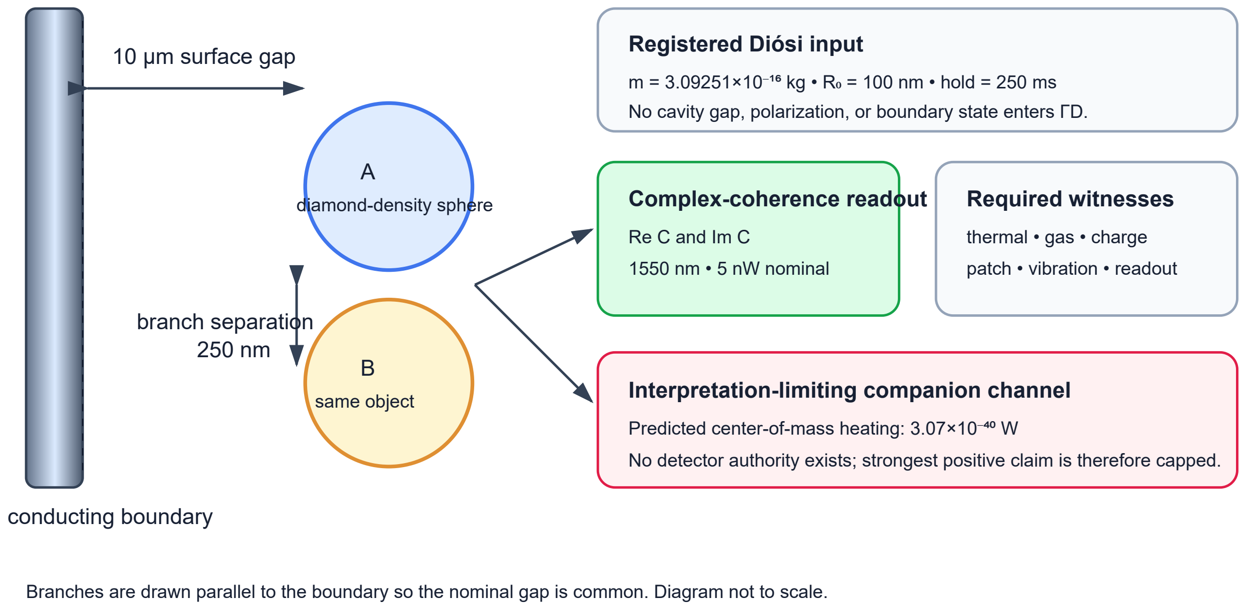

Figure 1. The boundary is an electromagnetic control, not a term in the registered Diósi generator. The two center-of-mass branches are separated parallel to the boundary so that the nominal surface distance is common to both branches. Environmental witnesses and a separate companion channel are required for interpretation.

Pilot and confirmatory timing logic

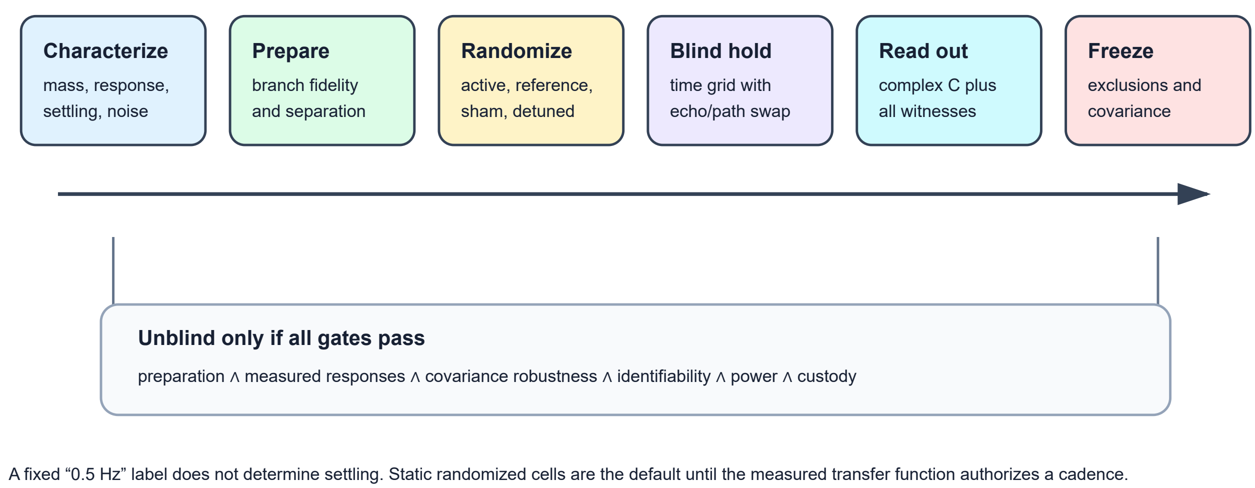

Figure 2. Preparation and branch verification precede the blinded hold. Path swap, echo, sham-switch, and detuned-boundary cells are randomized. Unblinding is permitted only after exclusions, response vectors, covariance ancestry, and analysis code are frozen.

1.3 Evidence tiers and present readiness

Every quantitative statement in this paper belongs to one of four evidence tiers:

Evidence tier	Meaning in this paper	Current state
software recovery	equations, ordering, provenance, and fail-closed behavior recover in synthetic fixtures	available
public-data component replay	a bounded analysis operation runs on a real external dataset	available separately; no cross-apparatus fusion
conditional apparatus forecast	a prediction under the selected synthetic apparatus and covariance assumptions	available, not empirical
same-apparatus measurement	jointly acquired preparation, response, covariance, coherence, and companion evidence	absent

The latest integrated gate therefore reads:

> **Protocol/software readiness: PASS. Subsystem commissioning: permitted. Integrated feasibility pilot: NOT AUTHORIZED. Same-apparatus authority packets: 0/8 ready. Measured collapse evidence: not ready. Collapse identification and manifold dynamics: blocked.**

The 2.908% contraction is presently a single leading-point sensitivity target, not yet a demonstrated test of the complete m, d, t law. The acquisition grid, statistical analysis plan, physical interpretation of a paired allocation unit, and apparatus-specific systematic budget must be closed by subsystem commissioning and the integrated feasibility pilot before the 1,028-unit estimate can become an actionable duration.

Three stages are deliberately distinct. **Subsystem commissioning**, which may proceed now, produces the eight authority packets and includes only low-statistics four-cell readout checks, not collapse-model scoring. An **integrated feasibility pilot** may begin only after all eight packets qualify; it tests the crossed design, response geometry, covariance, and analysis gates. A blinded **confirmatory acquisition** may begin only after that pilot passes and its protocol is publicly preregistered.

1.4 What expert review is being asked to decide

This draft is intended to elicit bounded decisions from specialists:

- **mesoscopic-state preparation:** identify or reject a platform capable of the registered mass, separation, hold, and recombination;
- **collapse theory:** assess the R_0 choice, composite mass-density map, external-bound recast, and companion relation;
- **quantum-gravity foundations:** assess the relational branch-state and equivalence-principle preflight, identify a defensible branch-correspondence construction, or explain why no such invariant construction is possible;
- **Casimir and macroscopic QED:** specify the selected specimen, measured spectra, as-built Green response, force-noise spectrum, and uncertainty;
- **relativistic phase and metrology:** close three-dimensional worldlines, echo/path-swap transfer, local gravity, and joint covariance;

- **statistics:** freeze the m, d, t allocation, likelihood, robustness family, stopping rule, and replication criterion.

The study should be judged as a model-specific design and subsystem- commissioning protocol seeking expert collaboration, not as an experiment-ready apparatus or a detection claim.

Reading order. Sections 2--4 define the tested theory, target state, and hypotheses. Section 5 maps every observable and ordinary-background channel into the complex-coherence analysis. Sections 6--9 define reach, pilot admission, collaboration packets, and inference rules. Detailed runtime chronology, superseded designs, and machine receipts are retained in the reproducibility supplement rather than used as scientific evidence.

2. Exact tested dynamics

2.1 Registered Gaussian-regularized Diósi model

The executable prediction is not a generic "Diósi–Penrose" rate. It is one named reduced-order implementation: a nondissipative Diósi mass-density master equation for a single effective particle whose mass-density operator is Gaussian-smeared over the physical length R0 [1–3]. In schematic form,

$$\frac{d\hat{\rho}}{dt} = -\frac{i}{\hbar}[\hat{H}, \hat{\rho}] - \frac{G}{2\hbar}\int d^3x\, d^3y\, \frac{[\hat{M}_{R_0}(\mathbf{x}), [\hat{M}_{R_0}(\mathbf{y}), \hat{\rho}]]}{|\mathbf{x}-\mathbf{y}|}. \tag{1}$$

The implementation freezes the convention, the mass representation, and R0. For two branches of the same effective Gaussian particle, with mass m and center separation d, the registered self-energy difference is

$$E_G(m, d, R_0) = Gm^2\left[\frac{1}{\sqrt{\pi}R_0} - \frac{\mathrm{erf}(d/2R_0)}{d}\right], \qquad \Gamma_{\mathrm{D}} = \frac{E_G}{\hbar}, \qquad \mathcal{V}_{\mathrm{D}}(t) = e^{-\Gamma_{\mathrm{D}} t}. \tag{2}$$

The d→0 limit is evaluated analytically. A Fourier-space Simpson quadrature is used only as a numerical cross-check; its softening parameter is not the physical cutoff.

For the leading synthetic commissioning point,

$$\begin{aligned} m &= 3.0925053 \times 10^{-16}\ \mathrm{kg}, \\ d &= 2.50 \times 10^{-7}\ \mathrm{m}, \\ R_0 &= 1.00 \times 10^{-7}\ \mathrm{m}, \\ E_G &= 1.2448784 \times 10^{-35}\ \mathrm{J}, \\ \Gamma_{\mathrm{D}} &= 1.1804586 \times 10^{-1}\ \mathrm{s}^{-1}, \\ \tau_{\mathrm{D}} &= 8.47128\ \mathrm{s}. \end{aligned} \tag{3}$$

At t=0.25 s,

$$1 - \mathcal{V}_{\mathrm{D}} = 1 - e^{-\Gamma_{\mathrm{D}} t} = 2.90803 \times 10^{-2} = 2.90803\%. \tag{4}$$

This is the registered effective-Gaussian forecast for the leading design. A second value is required because the mass-density representation is not yet an empirical authority. Transporting the existing representation envelope gives an exponent of 0.00435852, or 0.434903% loss at the same hold time. This is the lowest transported diagnostic forecast among the representations evaluated so far, not a physical lower bound: selected-specimen homogeneous, layered, coarse-grained, and atomistic maps remain uncomputed. Neither value is a total measured-visibility forecast.

2.1.1 From Schrödinger evolution to the measured residual

The experiment does not measure E_G directly. Its theoretical target is the off-diagonal center-of-mass density-matrix element $\rho_{AB}(t) = \langle A|\hat{\rho}(t)|B\rangle$, or its normalized form $c_{AB}(t) = \rho_{AB}(t)/\rho_{AB}(0)$. Section 5 defines the calibrated fringe estimator used to infer this quantity; the two are not identified without that readout map. For two Hamiltonian energy eigenstates, ordinary Schrödinger evolution gives

$$|\psi(t)\rangle = c_1 e^{-iE_1 t/\hbar}|E_1\rangle + c_2 e^{-iE_2 t/\hbar}|E_2\rangle, \qquad \rho_{12}(t) = c_1 c_2^* e^{-i(E_1-E_2)t/\hbar}.$$

Thus the beat frequency is $(E_1 - E_2)/\hbar$, while $|\rho_{12}(t)| = |c_1 c_2^*|$ remains constant in an ideal closed system. The energy variance $\sigma_H^2 = \langle H^2\rangle - \langle H\rangle^2$ describes the outcome spread of Hamiltonian-energy measurements; it is not the gravitational mass-density difference energy E_G. The localized apparatus branches $|A\rangle$ and $|B\rangle$ need not themselves be stationary energy eigenstates, so the experiment reconstructs their projected complex coherence rather than assuming a literal two-line atomic spectrum.

For the registered hypothesis lanes, the transparent scalar form is

$$c_{AB}(t) = e^{-i\Delta E_H t/\hbar} e^{-\chi_{\mathrm{env}}(t)} e^{-E_G t/\hbar}. \tag{5}$$

The Hamiltonian energy difference ΔE_H rotates phase. The ordinary open-system functional χ_{env} represents environmental contraction after the environment is traced out. The final factor is the additional nonunitary contraction of the frozen Diósi model. Equal units do not make these three objects interchangeable: in particular, a Schrödinger energy variance, a photon frequency, and the gravitational mass-density difference energy are not the same source term.

After ordinary loss has been estimated from blinded controls, a remaining scalar contraction can be expressed as

$$E_{\mathrm{D,eq}} = -\frac{\hbar}{t}\ln|c_{AB,\mathrm{residual}}(t)|\,. \tag{6}$$

Equation (6) is a conditional inference under the registered exponential model, not a calorimetric measurement of gravitational energy. The full whitened complex estimator in Eqs. (12)--(13), rather than this scalar summary, remains authoritative for distinguishing phase, loss, and correlated nuisance responses.

2.1.2 Constituent mass density changes the forecast

Equation (2) treats the apparatus as one effective Gaussian particle. Stage 4.2J also evaluates a homogeneous sphere of physical radius R, convolved with the identical Gaussian regularization R_0. Its radial Fourier form is

$$E_G^{\mathrm{sphere}} = \frac{2Gm^2}{\pi R_0}\int_0^\infty du\, e^{-u^2}\left[\frac{3(\sin q - q\cos q)}{q^3}\right]^2\left[1 - \frac{\sin(ud/R_0)}{ud/R_0}\right], \qquad q = \frac{uR}{R_0}. \tag{7}$$

For the earlier silica reference, the converged homogeneous-sphere diagnostic was 0.249896 of its effective-Gaussian energy. The subsequent representation audit widened the registered envelope to a factor of 6.77099. The bounded search transports that factor to the leading candidate rather than pretending that a diamond-density label supplies an atomistic density map. This yields the 0.434903% transported diagnostic benchmark quoted above. A direct homogeneous, layered, coarse-grained, and atomistic calculation for the selected specimen remains blocked pending provenance-bound density and coating inputs.

2.1.3 Cheap feasibility screens precede power

For an ideal equilibrium residual gas, the conservative screen uses

$$n = \frac{P}{k_B T}, \qquad \bar{v} = \sqrt{\frac{8 k_B T}{\pi m_g}}, \qquad \Gamma_{\text{coll}}^{\text{screen}} = n \bar{v}\, \pi R^2. \tag{8}$$

The earlier 2×10^{-11} Pa silica reference failed this screen. The leading design therefore registers 4 K and 10^{-15} Pa. Its transported transported QLBE proxy gives $\Gamma_{\text{gas}} = 8.64118 \times 10^{-4}\ \text{s}^{-1}$, or 0.007320 of the registered Diósi rate. This clears the synthetic one-tenth-Diósi gate, but it is not a measured vacuum receipt or a complete collisional decoherence kernel [13]. Species-resolved differential scattering, confinement, pressure calibration, and collision-veto performance must replace the proxy. The leading mass is also about 1.0955×10^6 times the 170 kDa cross-platform benchmark cited in Ref. [6], so state preparation remains the first independent hardware gate.

Hydrogen and Rydberg scales remain useful dimensional calibration checks. The registered E_G is about 5.71×10^{-18} of the Rydberg energy, but that ratio supplies neither a cavity coupling nor a collapse mechanism. It only states the scale of the residual that the coherence estimator would infer if the frozen Diósi model were selected after ordinary channels were excluded.

2.2 Penrose OR motivation, notation, and scope

Penrose's objective-reduction argument motivates an order-of-magnitude lifetime $\tau_{\text{OR}} \sim \hbar / E_G$ for incompatible mass distributions [4,5]. That heuristic and Eq. (2) share a gravitational self-energy scale, but they are not the same theory object:

Question	Penrose OR motivation	Registered Diósi runtime
Why consider finite branch lifetime?	incompatibility of alternative spacetime geometries	not supplied
Numerical lifetime scale	$\tau \sim \hbar / E_G$	$\Gamma = E_G / \hbar$
Stochastic master equation	not uniquely specified	specified
Gaussian cutoff R0	not fixed by OR	frozen at 100 nm
Diffusion/heating companion	not uniquely specified	specified
Casimir-boundary modifier	not supplied	absent

Consequently, an exclusion of Eq. (2) is not an exclusion of every Penrose-motivated theory. A match to Eq. (2) is not direct evidence for spacetime “snapping” or manifold instability.

2.2.1 A registered missing-theory program, not a new prediction

The equivalence-principle argument can be made more precise without pretending that it is already complete. Penrose's concern is not merely that a gravitational potential appears in a quantum Hamiltonian. Each superposed mass distribution would define a different freely falling or spacetime description, and general relativity supplies no automatic point-by-point identification of the events, clocks, and time translations belonging to those alternatives [4,34]. Competing quantum-reference-frame work also shows why equivalence plus superposition should not be presented as a theorem that uniquely implies collapse [35].

We have therefore registered a Stage-0 **relational branch-incompatibility candidate**. A branch is formally represented by its manifold, metric, quantum state, complete renormalized stress tensor, and physical reference system. The candidate then asks for four structures that Penrose's lifetime notation alone does not supply:

1. a relational rule identifying physical events and clocks across the two branches;
2. a gauge-independent incompatibility functional with energy units that recovers the Newtonian E_G expression;
3. a causal reduction dynamics with a survival distribution, Born probabilities, normalization, no-signalling, vacuum stability, and energy--momentum bookkeeping; and
4. a projection into both quadratures of measured complex coherence plus an independently testable companion or a sourced reason that none occurs.

In this program, E_I names the not-yet-supplied covariant branch- incompatibility functional. Its required stationary Newtonian recovery is $E_I^N \to E_G$, where E_G is the weak-field mass-density energy used in Eq. (2). Recovering E_G therefore checks a limit; it does not define E_I.

These requirements reflect known difficulties in relativistic reduction and semiclassical/stochastic-gravity programs [36--38]. The maintained preflight stops deterministically at `PCT_BRANCH_CORRESPONDENCE_MISSING`. It records $\tau_{\mathrm{OR}} \sim \hbar/E_G$ only as a heuristic scale, returns no numerical collapse rate, admits no held-out prediction vector, and makes no empirical validation claim. A schema-complete future definition would still remain Stage 0 until a separate source-backed calculator, recovery tests, and preregistration passed.

2.2.2 Stage-0.1 relational-correspondence benchmark

The first missing step now has an executable benchmark without being declared solved. In a static weak-field laboratory domain, a branch-blind apparatus clock and three apparatus fiducials define a shared relational label space B_{lab}. Two embeddings, $X_A : B_{\mathrm{lab}} \to \mathcal{M}_A$ and $X_B : B_{\mathrm{lab}} \to \mathcal{M}_B$, give the proposed local event correspondence $\varphi^{\mathrm{corr}}_{A\to B} = X_B \circ X_A^{-1} : X_A(B_{\mathrm{lab}}) \to X_B(B_{\mathrm{lab}})$. The superposed probe is not allowed to be the sole anchor because that could align away the displacement being tested. This is a local affine material-reference proxy, not a computed Fermi/radar chart or a preferred global map between arbitrary spacetimes [35,39--42].

The frozen benchmark requires identity recovery, an invertible map and inverse, branch-swap symmetry, invariance under independent rigid coordinate redescriptions, preservation of a known branch displacement, equality of the declared common-acceleration and common-potential inputs, agreement between two independently named reference sets, coverage of the declared $6R_0$ Gaussian finite-support proxy in every primary and alternate chart, and recovery of the Gaussian Newtonian target in Eq. (2). The exact named gate ledger in the content-addressed Stage-0.1 report is authoritative:

$$\mathcal{G}_{\mathrm{corr}} = \mathcal{G}_{\mathrm{id}} \wedge \mathcal{G}_{\mathrm{inv}} \wedge \mathcal{G}_{\mathrm{swap}} \wedge \mathcal{G}_{\mathrm{coord}} \wedge \mathcal{G}_{\mathrm{sensitivity}} \wedge \mathcal{G}_{\mathrm{common}} \wedge \mathcal{G}_{\mathrm{ref}} \wedge \mathcal{G}_{E_G}.$$

The displayed conjunction is a grouped mnemonic, not an alternate gate list: `G_ref` expands to the reference-subsystem, branch-map, declared-support, causal-order, alternate-reference, and reference-spread rows; `G_inv` expands to Jacobian and inverse-map rows; `G_coord` is rigid relabeling only; `G_common` is common-input equality only; and the authoritative ledger also contains its explicit output-policy row.

All synthetic gates pass. The pulled-back branch-center separation is 2.5000×10^{-7} m; the registered analytic target is $1.2448784 \times 10^{-35}$ J; and the independently integrated Fourier value is $1.2448786 \times 10^{-35}$ J, a relative discrepancy of 1.88×10^{-7}. The identity target and branch-swap residual are zero, and the independent coordinate-relabel residual is 3.56×10^{-22} m.

That is a software/theory recovery result. All five physical authority packets: measured worldlines, clock/pulse dictionary, as-built landmarks and tetrads, selected-specimen density support, and reference-choice covariance, remain `not_ready`. The scientific standing therefore stops at `PRC_REFERENCE_RECEIPTS_MISSING`, and the parent Stage-0 candidate remains at `PCT_BRANCH_CORRESPONDENCE_MISSING`. The frozen v1 schema cannot be filled in place to claim readiness; measured packet admission requires a versioned successor with path, hash, custody, calibration, and uncertainty validation.

Passing Stage-0.1 shows that one proposed local correspondence survives frozen local affine and rigid-relabeling tests; it does not select a unique physical correspondence, establish a covariant incompatibility functional, or generate a lifetime, collapse rate, coherence law, Casimir modifier, or model-comparison row. The content-addressed gate ledger is in the Stage-0.1 report. The rigid-relabeling pass is not arbitrary diffeomorphism covariance, and common-input equality is not the still-blocked full equivalence-principle recovery obligation.

The boundary policy is deliberately independent. At matched branch mass-density and trajectory histories, changing the Casimir boundary does not modify this intrinsic candidate. Any proposed boundary modifier is a different model and must pass the existing tensor/noise/retarded-response manifold registry. Thus the experiment remains two-axis: the mass-separation-time campaign tests the registered Diósi model today and could later test a completed Penrose candidate; the four-cell estimator separately tests boundary-superposition nonfactorization.

The full obligation, nonbridge, falsifier, and outcome ledger is preserved in the Stage-0 candidate preflight report.

2.3 Mass-density representation is part of the hypothesis

The leading object's geometric radius is 276.3 nm, but Eq. (2) treats its center of mass as one Gaussian-smeared effective particle. This is not equivalent to resolving a selected specimen's atomic or layered density. The current representation audit is:

Representation of the same apparatus	Current standing	Consequence
single effective Gaussian particle	executable; Eq. (2)	2.90803% leading-design forecast
transported representation envelope	executable diagnostic benchmark	0.434903% lowest transported result evaluated so far; not a selected-specimen bound or map
homogeneous selected sphere	not yet recomputed for the selected material identity	bulk-profile dependence unresolved
coated or layered selected sphere	not computed	coating dependence unknown

Representation of the same apparatus	Current standing	Consequence
coarse-grained or atomistic selected density	no admissible density provenance	granularity dependence unknown
R0 sweep with converged selected representations	sensitivity-only calculations exist	parameter stability unresolved

Therefore this article proposes a test of the effective-particle Diósi model, not a representation-independent test of the broader collapse-model family. A revised confirmatory analysis must either demonstrate stability across physically defensible mass representations or preserve that narrower claim in its title and conclusion.

3. Leading Synthetic Commissioning Design

The following manifest is the current target for measured subsystem commissioning. It supersedes earlier design assumptions for forward planning, but it does not claim an as-built apparatus or physical feasibility.

Field	Leading design value	Evidence role
identity	`stage4_2m_candidate_002`	frozen synthetic search identity
material and geometry	diamond-density sphere	bounded-search assumption; specimen not selected
radius	276.302 nm	derived from frozen mass/material identity
mass	$3.0925053 \times 10^{-16}\ \mathrm{kg}$ $1.86235 \times 10^{11}\ \mathrm{Da}$	frozen model input
branch separation	250 nm, parallel to boundary	frozen model input
principal hold time	250 ms	frozen model input
hold-time requirement	include prespecified shorter and longer cells	identifiability check
nominal surface gap	10 µm, common to both branches	boundary-control input
finite plate size	80 µm × 80 µm	transported finite-geometry input
boundary program	static randomized cells; nominal cadence label 0.5 Hz	transfer function unmeasured
primary sequence	Ramsey-type branch preparation/recombination	design assumption
control sequences	identical-branch/sham splitter, echo, path swap, sham switch, detuned boundary	required nuisance and interaction discrimination
temperature	4 K	design target, unmeasured in integrated apparatus
pressure	$1 \times 10^{-15}\ \mathrm{Pa}$	design target, unmeasured in integrated apparatus
registered echo residual	1×10^{-4}	synthetic control assumption
vibration target	$5 \times 10^{-10}\ \mathrm{m\,s^{-2}\,Hz^{-1/2}}$	design target
readout	1550 nm, 5 nW nominal	design assumption
polarization	circular-control pair plus linear-basis checks	electromagnetic witness
companion channel	center-of-mass energy increase	detector authority absent
settling rule	to be measured and frozen from transfer-function data	not inherited from old design

The “0.5 Hz” value is a configuration label, not permission to assume that a two-second cycle is quasistatic. No continuous modulation or ten-second settling rule is admitted until the measured boundary-to-apparatus transfer function defines a compatible sequence. Static randomized cells are the default pilot implementation.

3.1 Design selection status

The current diamond-density target follows two explicit rejections: the first candidate was nuisance-collinear, and the later silica parent failed the combined gas, phase, and preparation screens. The complete stage-by-stage crosswalk is preserved in Section 8.1 of the reproducibility supplement. “Frozen” means immutable within a particular run; it does not make a superseded run the current proposal. Only the manifest above is used for forward commissioning.

3.2 The dominant scale gap

The leading mass is approximately 1.86235×10^{11} Da. Matter-wave interference has recently been demonstrated for sodium clusters above 170,000 Da [6]. The proposed object is therefore about 1.0955×10^{6} times more massive than that free-particle interference benchmark. The comparison is not a universal impossibility theorem—trapped mechanical systems use different state-preparation strategies—but it prevents the manuscript from treating preparation as an ordinary checklist item.

State preparation is the first physical go/no-go:

1. verify the object's mass and internal-state distribution;
2. demonstrate coherent branch preparation at the registered separation;
3. verify the 250 ms branch hold without conditioning on successful recombination;
4. measure visibility and phase stability over the full control schedule;
5. show that the preparation procedure does not itself generate the fitted nuisance signature.

Until these steps are demonstrated, the leading 1,028-window requirement and 1,600-window planning ceiling are not actionable acquisition durations. The older 542-unit value remains only a parent-design record.

3.3 Platform position and required preparation receipt

This study is presently platform-agnostic. “Ramsey-type preparation/recombination” identifies the required logical sequence; it is not a demonstrated preparation Hamiltonian or a selected hardware implementation. The bounded preparation-scale screen used in the apparatus search was a mass, geometry, and separation filter. It supplies no evidence that the leading state can be prepared. The diamond sphere is therefore a target-state specification awaiting a platform receipt, not a build-ready apparatus.

A qualifying receipt must provide the preparation/control Hamiltonian and the trap, cooling, splitting, hold, recombination, and readout sequence; measured mass and internal state; three-dimensional packet centers and covariance; packet width, overlap, momentum difference, and trajectory; preparation and recombination fidelity including failed attempts without success-conditioned postselection; survival during the 250 ms hold; preparation-induced heating, charge, magnetic, optical, vibration, and boundary-correlated responses; and the sham, echo, path-swap, cycle-time, and duty-time characterization.

If no platform demonstrates the registered state while satisfying packet equivalence and covariance gates, the candidate is rejected and its power and window forecasts remain non-actionable. That is an apparatus no-go and carries no inference about the Diósi model.

3.4 Operational status of the m, d, t scaling test

The leading manifest defines a point prediction at $m = 3.0925053 \times 10^{-16}$ kg, $d = 250$ nm, and $t = 250$ ms. The 2.908% forecast can test pointwise compatibility with Eq. (2), but cannot by itself establish the complete mass--separation--hold-time dependence. The superseded silica grid is not transported to the diamond candidate.

Before confirmatory freeze, pilot capability measurements must populate and power a current-platform crossed grid. The existing hold-time contract requires a zero-time intercept, at least four distinct hold times, and a positive-time span of at least a factor of four. The exact current-platform times, mass/density cohorts, nonzero separations, allocation, and hierarchical model remain `not_ready`. Mass/material changes are between-object effects, not within-object knobs, and each cohort requires its own geometry, density, charge, polarizability, surface-response, and gas-scattering characterization. Separation cells must preserve the common boundary distance and measured three-dimensional branch geometry.

If an identifiable crossed grid cannot be realized, the experiment may report only pointwise compatibility or exclusion at the cells actually measured. It must not claim confirmation of the m, d, t scaling law.

4. Hypotheses separated before data

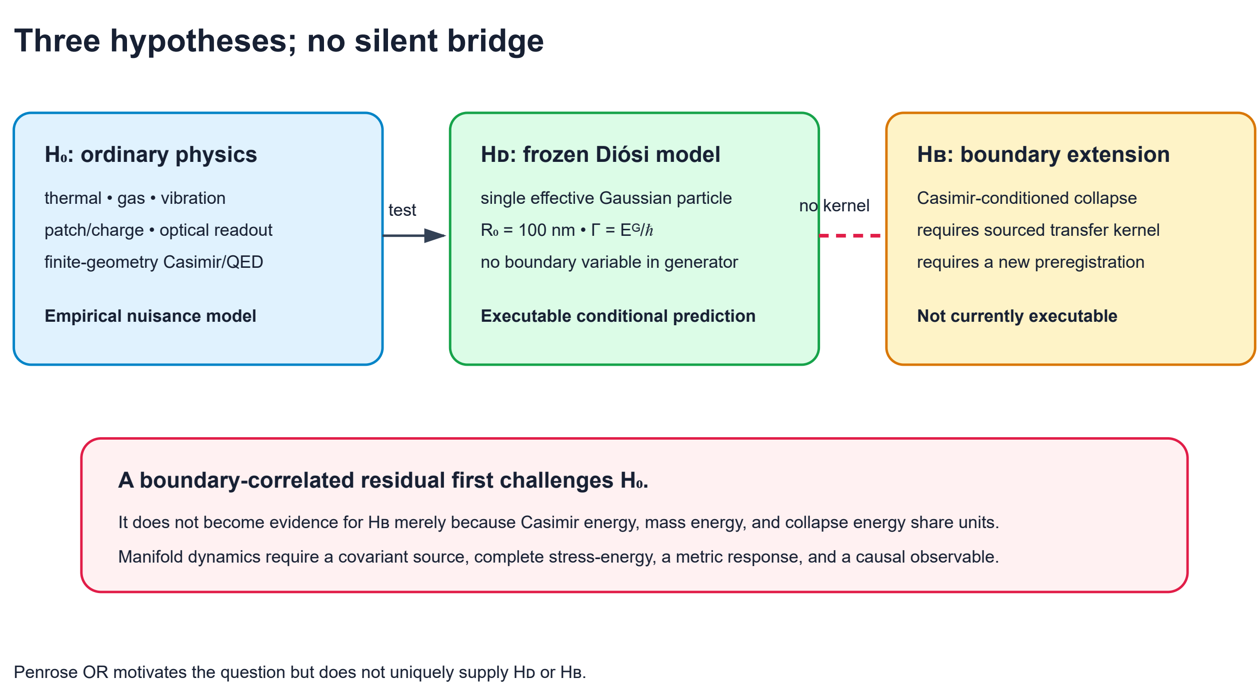

Figure 3. The ordinary-physics null, the frozen effective-particle Diósi generator, and a possible boundary-to-collapse extension are separate hypotheses. A Casimir-correlated residual cannot move from the left branch to the right branch without a registered transfer kernel and new data.

4.1 Ordinary-physics null, H0

H0 contains all registered ordinary mechanisms that can alter complex coherence:

- blackbody emission, absorption, scattering, and temperature drift;
- residual-gas collision and recoil;
- patch potentials, charge, dielectric response, and electromagnetic forces;
- vibration, tilt, support motion, and clock error;
- optical back-action, detector self-noise, and analysis leakage;
- boundary switching, material hysteresis, and finite-geometry Casimir/QED response.

The nuisance model is empirical. Textbook formulas provide priors and dimensional checks, not a substitute for measured response vectors.

4.2 Frozen Diósi hypothesis, HD

HD adds Eq. (2) to H0 with the mass, separation, R0, and hold-time law frozen before confirmatory data. The Casimir gap and modulation state do not enter the collapse generator. Under complete joint-system equivalence, standard HD therefore predicts the same collapse term in boundary-active and boundary-reference cells.

4.3 Lane C: boundary-conditioned extension, HB

HB is a future model slot in which a Casimir boundary changes collapse or coherence through an explicitly sourced transfer kernel. No such kernel is registered. The ideal Casimir energy [11],

$$\frac{E_\mathrm{C}}{A} = -\frac{\pi^2\hbar c}{720a^3}, \qquad P_\mathrm{C} = -\frac{\pi^2\hbar c}{240a^4}, \tag{9}$$

does not define a map from plate gap a to Γ_D. Real-material Lifshitz theory [12], finite geometry, temperature, patches, roughness, and nonequilibrium response belong first in H0.

4.4 Manifold-response hypothesis

The motivating manifold statement is a research question, not an admitted mechanism: if alternative mass configurations source incompatible gravitational geometries, a collapse-like lifetime might scale with a gravitational self-energy. The present nonrelativistic master equation tests one phenomenological realization of that scale. A spacetime claim would additionally require a covariant source, a complete apparatus stress-energy tensor, a specified dynamical metric response, and a causal observable. Those objects are absent.

4.5 Compton-frequency non-bridge

Rest energy may be written as a frequency, $\nu_\mathrm{C} = mc^2/h$, and the collapse energy as $\nu_G = E_G/h$. Atomic, QED, Higgs, blackbody, Zeeman, Stark, Maxwell, and gravitational benchmarks use compatible energy and frequency units. That compatibility is valuable for calibration and unit recovery but does not create a coupling between cavity modes and collapse:

$$\mathcal{K}_{\text{boundary}\to\text{branch/coherence}} \quad \text{is not registered.} \tag{10}$$

The same rule applies to light cones, spinors, gravitational waves, Jeans instability, Schwarzschild radii, and mass–energy equivalence. They validate parts of relativistic or quantum notation; they do not supply Eq. (10).

5. Complex-coherence estimator and identifiability

The primary datum is not a fitted scalar decay constant. After a calibrated readout maps fringe quadratures to the branch basis, each control cell supplies the complex fringe-coherence estimator

$$C_k(t) = \langle e^{i\phi} \rangle_k = \operatorname{Re} C_k + i \operatorname{Im} C_k. \tag{11}$$

Here C_k is the calibrated fringe observable before zero-hold normalization; it is related to ρ_{AB} by the registered readout model. Section 5.1 defines $\bar{C}_k(t) = C_k(t)/C_k(0)$, which estimates the normalized theoretical coherence c_{AB}. Real and imaginary components retain phase rotations that a visibility-only analysis would discard. Recovery after independent phase conditioning, echo, or path swap diagnoses reversible dephasing rather than intrinsic contraction. A non-exponential line shape alone is not evidence for collapse: prespecified ordinary and Diósi line shapes must be compared on the complete hold-time grid. Let y be the stacked real vector of all components, X the matrix of registered nuisance signatures, sD the frozen Diósi signature, and Σ the block covariance. Whitening gives

$$\tilde{y} = L^{-1}y, \qquad \tilde{X} = L^{-1}X, \qquad \tilde{s}_{\mathrm{D}} = L^{-1}s_{\mathrm{D}}, \qquad LL^{\mathsf{T}} = \Sigma. \tag{12}$$

The collapse-sensitive component is the part orthogonal to the whitened nuisance span:

$$s_{\perp} = (I - P_{\tilde{X}})\tilde{s}_{\mathrm{D}}. \tag{13}$$

Integrated ordinary-physics authority map. The response-function discipline used in mature interferometer and force-metrology proposals is applied here as a fail-closed requirement [24--29]. Each channel must be propagated into the same complex-coherence space and joint covariance used by Eqs. (12)--(13).

Channel	Transfer to the measured datum	Reversal or witness	Required authority	Current standing
state preparation, recombination, and specimen identity	packet mismatch, phase, contrast loss, or wrong mass-density representation	sham split, path swap, tomography, failed-attempt ledger, mass/coating assay	measured packet/worldline and specimen-density receipt for every boundary state	absent
material and electromagnetic response	finite-geometry QED phase, force gradient, heating, and FDT contraction	gap, polarization, detuning, impedance state, independent solver	as-built geometry, measured spectra, Green tensor, convergence, and covariance	synthetic fixture only
gas and thermal environment	collisional and radiative contraction, drift, and heating	species/pressure/temperature monitors and collision veto	species-resolved collision kernel and measured thermal transfer	absent
gravity, kinematics, and control	signed proper-time, gradient, Sagnac, laser, and timing phase	branch-vector reversal, echo, path swap, local-mass survey	measured 3D worldlines and joint phase covariance	synthetic budget only
magnetic boundary toggle	electromagnetic phase/loss, trap transfer, vortex and hysteresis response	fixed-temperature field sham and pickup channels	measured complex impedance and state-conditioned Maxwell response	candidate strategy only
readout and covariance	phase mixing, apparent visibility loss, and estimator bias	raw I/Q retention, blind labels, train/holdout split	calibrated transfer, attrition record, positive-definite frozen covariance	absent

Channel	Transfer to the measured datum	Reversal or witness	Required authority	Current standing
same-generator companion	diffusion/heating or another derived channel	independent sensor and cross-covariance audit	instrument-level prediction using the identical frozen parameters	no detector authority
external bound	admissibility of the exact parameter convention	independent published analysis	exact variant and composite-particle recast	scalar screen only

An ordinary channel without a same-apparatus transfer function and covariance is treated as unbounded, not as zero. No synthetic ordinary-physics value is eligible for subtraction from measured coherence. If the permitted response of any open channel destroys the rank, cosine, conditioning, phase, or covariance gates, confirmatory acquisition is not admitted.

5.1 Boundary--branch interaction diagnostic

The full whitened estimator remains primary, but the boundary question has a transparent four-cell projection. Let $\beta = 0, 1$ denote reference and active boundary states, and let $q = 0, 1$ denote a measured identical-branch (or prespecified sham-split) control and the separated material superposition. Normalize each cell without discarding phase,

$$\bar{C}_{\beta q}(t) = \frac{C_{\beta q}(t)}{C_{\beta q}(0)}.$$

The four-cell diagnostic freezes the complex cross-ratio

$$R_4 = \frac{\bar{C}_{11}\bar{C}_{00}}{\bar{C}_{01}\bar{C}_{10}}, \qquad I_4 = -\ln|R_4|, \qquad \Phi_4 = \arg R_4. \tag{14}$$

The registered ordinary-physics prediction is not assumed to factorize perfectly. Its measured response model supplies $R_{4,0}$, and the corrected interaction is

$$R_{4,\mathrm{corr}} = \frac{R_{4,\mathrm{obs}}}{R_{4,0}}, \quad I_{4,\mathrm{corr}} = I_{4,\mathrm{obs}} - I_{4,0}, \quad \Phi_{4,\mathrm{corr}} = \mathrm{wrap}(\Phi_{4,\mathrm{obs}} - \Phi_{4,0}). \tag{15}$$

Standard H_D multiplies both separated-branch boundary cells by the same $e^{-\int \Gamma_\mathrm{D}[d(t)]dt}$, provided the registered mass-density and branch-separation histories are matched between boundary states. That factor cancels from Eq. (14), so this statistic is not the primary standard-Diósi test. It asks the separate question: does the boundary change branch-dependent coherence? A corrected $R_{4,\mathrm{corr}} \neq 1$, equivalently nonzero $I_{4,\mathrm{corr}}$ or $\Phi_{4,\mathrm{corr}}$, first challenges H_0 or complete-joint-system equivalence. It cannot identify a Casimir-to-collapse mechanism without the separate kernel prohibited by Eq. (10).

For log-loss cells $Y = (Y_{00}, Y_{01}, Y_{10}, Y_{11})$, the elementary contrast is $c^\mathsf{T} Y$ with $c = (1, -1, -1, 1)^\mathsf{T}$ and variance $c^\mathsf{T} \Sigma_Y c$. The synthetic recovery reproduces this as the saturated four-cell special case of covariance-weighted projection; Eq. (12), with all controls and quadratures retained, remains the general estimator. If any normalized coherence is outside the registered log-coverage domain, Eq. (14) is withheld and the raw complex cells remain authoritative.

The same campaign replaces an opaque wave-packet receipt with explicit center-of-mass custody:

$$\mathbf{d} = \mathbf{x}_B - \mathbf{x}_A, \qquad \sigma_\mathrm{CM} = \sqrt{\frac{\mathrm{tr}\,\Sigma_\mathrm{CM}}{3}}, \qquad \{R_\mathrm{sphere}, R_0, \sigma_\mathrm{CM}\} \quad \text{have distinct physical and model roles.} \tag{16}$$

Centers, packet covariance, overlap, momentum difference, separation uncertainty, hold jitter, preparation fidelity, trajectory, and tomography provenance must agree across boundary states within frozen tolerances. The maintained fixture exercises this contract synthetically with $\sigma_{\mathrm{CM}} = 10\ \mathrm{nm}$; it is not a prepared-state measurement. In that boundary-independent-Diósi recovery, $I_{4,\mathrm{corr}} = 1.11 \times 10^{-16}$, $\Phi_{4,\mathrm{corr}} = -9.30 \times 10^{-19}\ \mathrm{rad}$, and the maximum interaction significance is 3.85×10^{-13}. An adversarial fixture recovers an injected (0.002) loss interaction and $(0.004\ \mathrm{rad})$ phase interaction, while low coherence, packet mismatch, non-positive covariance, and boundary-dependent Diósi attenuation fail closed. These are software-recovery results only; the four cells, packet metrology, and ordinary response are not measured.

The prespecified nominal design gates are

$$\max_j |\cos(\tilde{s}_{\mathrm{D}}, \tilde{x}_j)| \leq 0.97, \quad \kappa(G_{\mathrm{norm}}) \leq 100, \quad \mathrm{power} \geq 0.80, \quad \mathrm{FPR} \leq 0.05. \tag{17}$$

These are design-admission gates, not proof that the modeled nuisance set is complete.

5.2 Rejected design

The initial synthetic design produced

Diagnostic	Result	Gate
maximum absolute whitened cosine	0.999977	fail
normalized Gram condition number	179,104	fail
acquisition power	withheld	not meaningful

The correct result was an apparatus-design no-go. It was not a null result on the Diósi model.

5.3 Parent synthetic redesign

The parent synthetic redesign produced

Diagnostic	Nominal synthetic value	Gate
maximum absolute whitened cosine	0.717724	pass
normalized Gram condition number	6.53169	pass
forecast power	0.997858	pass in nominal world
nominal paired allocation units	542	dimensionless synthetic result
planned allocation ceiling	1,600	not yet an acquisition plan

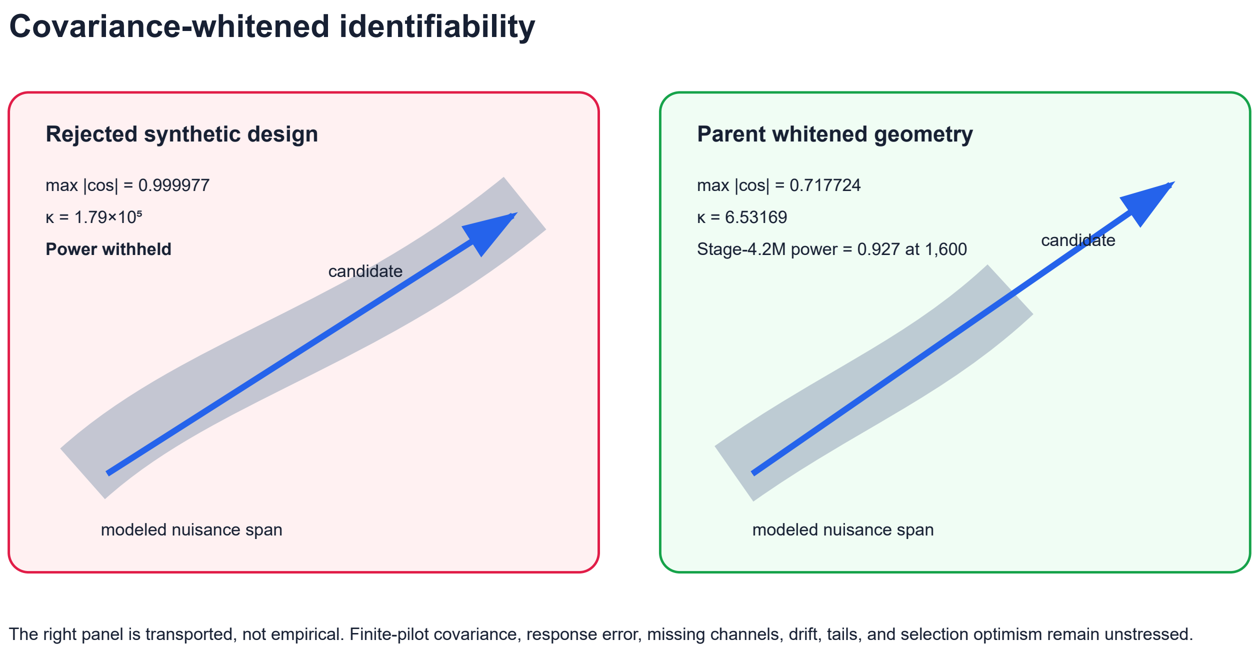

Figure 4. The first design's candidate direction lies almost inside the nuisance span. The parent redesign establishes the nominal whitened geometry later transported by the bounded search. Both panels remain conditional on synthetic response vectors and an assumed covariance.

5.4 Selected bounded design

The downstream search evaluates 200 bounded configurations under the frozen Diósi law and the same registered whitened geometry. Three survive all bounded synthetic search filters; none has physical companion authority. The leading point is:

Diagnostic	Leading synthetic value	Gate
maximum absolute whitened cosine	0.717724	pass; transported geometry
normalized Gram condition number	6.53169	pass; transported geometry
effective-Gaussian visibility loss at 250 ms	2.90803%	conditional named-model forecast
lowest transported density diagnostic at 250 ms	0.434903%	representation benchmark; not a physical floor
gas-to-Diósi proxy ratio	0.007320	conditional synthetic screen
transported electromagnetic phase-jitter surrogate	1.109×10^{-8} rad	conditional; excludes gravity/worldline terms
integrated proper-time/control phase standard uncertainty	0.018007 rad	conditional on unmeasured echo and worldline assumptions
companion scaling surrogate	8.668	ranking arithmetic only; no detector authority
required paired allocation units	1,028	dimensionless planning value; no duty-cycle map
forecast power at 1,600 windows	0.927386	conditional synthetic pass

This table, not the 542-window parent result, is the current synthetic allocation forecast. The value 8.668 transports an assumed reference SNR by mass and readout-efficiency scaling; it is not derived from the predicted 3.07013×10^{-40} W heating power and a detector-noise model. The point therefore does not pass a physical companion-feasibility gate. It remains a subsystem-commissioning target because its response, covariance, gas, phase, preparation, and companion inputs are transported surrogates rather than measurements.

5.5 Material-resolved ordinary complex-coherence null

The material/Green/FDT analysis module replaces the transported electromagnetic scalar with an executable ordinary-response chain for the leading design. A passive specimen loss table is converted to $\epsilon(i\xi)$, a finite-geometry Green table supplies the mean branch potential, and a two-sided fluctuation--dissipation spectrum supplies phase/loss covariance [30]. The transparent ordinary prediction is

$$C_{0,\beta}(t) = C(0)\exp[i\Phi_{\mathrm{EM},\beta}(t) - \chi_{0,\beta}(t)]\,. \tag{18}$$

The boundary-by-superposition control is represented by the normalized four-cell ratio

$$R_4 = \frac{\bar{C}_{\text{active,separated}}\bar{C}_{\text{reference,compact}}}{\bar{C}_{\text{active,compact}}\bar{C}_{\text{reference,separated}}}, \qquad \Phi_4 = \arg R_4, \qquad I_4 = -\ln|R_4|. \tag{19}$$

This is the explicit form of the comparison already defined in Section 5.1; each barred cell is normalized to its own zero-hold reference. A corrected $R_{4,\mathrm{corr}} \neq 1$ challenges factorization of the boundary and branch responses; it does not by itself identify collapse. The boundary-independent registered Diósi factor is identical in active and reference cells and therefore cancels from this interaction statistic when their mass-density and separation histories match. The separate Diósi exponents remain the mass--separation--time comparator for the primary coherence analysis; the module does not add them to the ordinary Green/FDT exponent and supplies no Casimir-to-collapse kernel.

The current synthetic fixture recovers $\Phi_4 = 0.01999999999$ rad and $I_4 = 1.34165 \times 10^{-6}$, with propagated phase uncertainty 6.30×10^{-5} rad. It also recovers zero response at zero coupling and infinite distance and a Stefan--Boltzmann relative error of 1.38×10^{-14}. These values validate executable bookkeeping only. Measured evidence and ordinary-null authority remain `not_ready`; residual attribution, collapse identification, and manifold dynamics remain `blocked`.

5.6 Public-data component validation

The public-data component campaign asks a deliberately narrower question than the proposed experiment: can the registered analysis code recover its constituent operations from real, open measurements? Four independent archives are assigned one role each.

Public record	Recovered operation	Result	Claim ceiling
sodium-cluster interference [6,31]	complex fringe coefficient	95 scans; alternating-split mean Mahalanobis squared 0.01143	measured matter-wave fringe reconstruction only
superconducting drum [21,32]	paired nonlinear boundary/drive response	108 traces; RMS up/down centroid shift 7.358 kHz	measured boundary-response replay only
LISA Pathfinder [33]	multichannel covariance and held-out residual	421,912 active rows; 16 channels; shrunk condition numbers 3.323 and 3.338	classical covariance/re-entry replay only

Public record	Recovered operation	Result	Claim ceiling
Gran Sasso underground spectrum [8]	external Diósi-bound source record	4,000-bin spectrum and 140-bin data/simulation comparison authenticated	parameter-free bound authority only

For the sodium record, the importer reconstructs the first normalized Fourier coefficient

$$S_\ell = \frac{1}{N}\sum_{j=0}^{N-1} n_j e^{-2\pi i\ell j/N}, \qquad \widetilde{C}_1 = 2\frac{S_1}{S_0} = V e^{i\phi}. \tag{20}$$

Here n_j is the binned detected count and the registered finite-bin phase correction is applied after the transform. The symbol $\widetilde{C}_1$ is a fringe-transfer observable. It is not identified with $\rho_{AB} = \langle A|\hat{\rho}|B\rangle$ for the proposed diamond sphere.

The LISA replay standardizes each channel on the training half, forms a regularized covariance, and evaluates a frozen linear residual on held-out windows:

$$\widehat{\Sigma}_{\mathrm{tr}} = \mathrm{Cov}(\mathbf{z}_{\mathrm{tr}}), \qquad \mathbf{r}_{\mathrm{ho}} = \mathbf{z}_{1,\mathrm{ho}} - \mathbf{X}_{\mathrm{ho}}\widehat{\boldsymbol{\beta}}_{\mathrm{tr}}. \tag{21}$$

This exercises covariance ancestry and held-out residual handling on a real multichannel physical record; it does not validate those operations for the proposed apparatus, quantum coherence, Casimir subtraction, or collapse. The superconducting-drum replay likewise establishes a measured paired spectral response but does not independently identify the force or calibrate this apparatus. The Gran Sasso replay authenticates the parameter-free exclusion record, but the registered $R_0 = 100$ nm comparator remains `not_adjudicated` until a variant-matched recast is supplied.

The separation rule is absolute: the four archives have different apparatuses, state spaces, transfer functions, and covariance ancestry. The campaign therefore constructs no shared likelihood, cross-apparatus covariance, transported residual, or observable bridge. Its component replay passes while `measured_evidence` and `joint_protocol_validation` remain `not_ready`, and collapse identification and manifold dynamics remain `blocked`.

5.7 Proper time, worldlines, and the ordinary phase budget

The Schrödinger Hamiltonian baseline must include the phase generated by the actual branch worldlines. In a stationary weak field, the branch difference is

$$\Delta\tau = \int_0^T \left[\frac{\Phi_A(t) - \Phi_B(t)}{c^2} - \frac{v_A^2(t) - v_B^2(t)}{2c^2}\right] dt, \tag{22}$$

and the corresponding propagation phase follows from the branch action:

$$\Delta\phi_{\mathrm{prop}} = -\frac{mc^2}{\hbar}\Delta\tau. \tag{23}$$

This is a physical transfer law from proper time to an ordinary **signed unitary phase**. It is not the unsupported Compton-frequency bridge rejected elsewhere in this paper: the experiment does not claim that the sphere exposes a directly readable Compton clock, and Equations (22)--(23) do not terminate in a collapse rate [14,15].

Spatial separation alone is insufficient. If the branches sample equal potentials and equal squared velocities, then $\Delta\tau = \Delta\phi_{\mathrm{prop}} = 0$ even when their centers are distinct.

For the leading apparatus, the registered separation vector is nominally horizontal with respect to local gravity. Its linear Earth-potential difference is therefore zero. The fully vertical reference below is a worldline-propagation sensitivity scale, not a standalone closed-interferometer observable:

$$\frac{\Delta\tau}{T} = \frac{gd}{c^2} = 2.72784 \times 10^{-23}, \qquad \Delta\tau = 6.81961 \times 10^{-24}\ \mathrm{s},$$

$$|\Delta\phi_g| = \frac{mgdT}{\hbar} = 1.79736 \times 10^{12}\ \mathrm{rad}.$$

Consequently, the declared 10^{-10} rad vertical-tilt standard uncertainty would map to about 180 rad before cancellation. The current calculation transports a 10^{-4} signed-phase echo residual and adds a small frequency-binned tilt response, giving an Earth-tilt phase-standard-uncertainty component of 0.017979 rad. Gravity-gradient, balanced local-mass, kinematic, Earth-rotation/Sagnac, laboratory-clock, control-phase, and electromagnetic phase-standard-uncertainty components are then added in quadrature, yielding $\sigma_{\phi,\mathrm{total}} = 0.018007$ rad. This is below the frozen 0.034644 rad design limit by 0.016638 rad. The margin is useful but not generous: without the assumed echo rejection the design fails by orders of magnitude.

A measurable closed-interferometer phase also contains the preparation, trap/support, pulse, recombination, endpoint, separation, and readout actions. The worldline calculation is therefore an admission screen until a complete apparatus sensitivity function or action calculation closes those terms.

Internal energy supplies a separate ordinary reduced-state bound:

$$C_{\mathrm{int}}(\Delta\tau) = \mathrm{Tr}\left[\rho_{\mathrm{int}} e^{-iH_{\mathrm{int}}\Delta\tau/\hbar}\right], \qquad |C_{\mathrm{int}}| \simeq \exp\left[-\frac{\mathrm{Var}(H_{\mathrm{int}})\Delta\tau^2}{2\hbar^2}\right]. \tag{24}$$

The synthetic variance screen is negligible, but no specimen heat-capacity or internal-spectrum receipt exists. It therefore remains an ordinary-background requirement rather than evidence. Echo and path swap are applied only to signed ordinary phases. The positive frozen Diósi exponent is unchanged only when those controls preserve the registered mass-density and separation history; otherwise $\int \Gamma_{\mathrm{D}}[d(t)]dt$ must be recomputed. It is never echo-scaled or added to the phase covariance [16,17]. Optical-clock redshift measurements establish the underlying proper-time metrology principle but do not supply the apparatus's matter-wave worldlines or covariance [18].

The proper-time module passes its software recoveries and synthetic total-phase screen. Measured three-dimensional worldlines, the local gravity-gradient map, the as-built local-mass CAD, tilt spectra, frequency-dependent echo transfer, clock/control covariance, and internal-energy variance remain `not_ready`. Measured evidence remains `not_ready`; collapse identification and manifold dynamics remain `blocked`; physical viability remains `not_evaluated`; and no proper-time-to-collapse or Casimir-to-collapse bridge is registered.

5.8 Superconducting boundary control: bridge and nonbridges

A superconducting boundary is useful here as an **ordinary-response control**, not as a proposed origin of Diósi collapse. For a charged condensate, London screening gives

$$\lambda_L^{-2} = \mu_0 \frac{n_s(q^*)^2}{m^*}, \qquad m_{\gamma,\mathrm{eff}} = \frac{\hbar}{\lambda_L c}. \tag{25}$$

The second quantity is an in-medium electromagnetic mass scale. It is not a vacuum photon rest mass and does not identify the Standard-Model Higgs field. Likewise, zero DC resistance does not imply zero finite-frequency impedance:

$$Z_s(\omega) = R_s(\omega) + iX_s(\omega), \qquad X_s(\omega) \simeq \mu_0 \omega \lambda_L \tag{26}$$

in the declared London-limit recovery. The experimentally meaningful chain is $Z_s, \sigma, \epsilon \to \mathbf{G}_{\mathrm{EM}} \to (\Phi_{\mathrm{EM}}, \chi_{\mathrm{EM}}) \to C$. That is an ordinary observable bridge into the ordinary Green/FDT null. It is not a condensate-to-collapse bridge [19--23].

For software recovery, the control analysis freezes a linearized weighted impedance- contrast surrogate and requires it to reproduce every supplied $(\Delta\Phi_{\mathrm{EM}}, \Delta\chi_{\mathrm{EM}})$ point. That numerical transfer passes, but it is explicitly a synthetic stand-in for the missing specimen-specific, finite-geometry Maxwell/Green calculation.

Writing the normal and superconducting boundary states as N and S,

$$C_\beta(t) = C(0) e^{i\Phi_{\mathrm{EM},\beta}(t) - \chi_{\mathrm{EM},\beta}(t) - \Gamma_D t}, \qquad \frac{C_S}{C_N} = e^{i(\Phi_S - \Phi_N) - (\chi_S - \chi_N)}. \tag{27}$$

The frozen boundary-independent Diósi factor cancels. This ratio therefore tests ordinary boundary response or an additional boundary-conditioned interaction; it cannot replace the primary mass--separation--hold-time Diósi contrast. The synthetic fixture recovers that cancellation to 2.22×10^{-16} in its synthetic fixture.

The bounded strategy result is deliberately asymmetric. Crossing T_c fails because the thermal nuisance is collinear with the intended contrast. A matched static pair fails its SNR and fabrication-degeneracy gates. A fixed-temperature magnetic toggle is the only synthetic candidate, with contrast SNR 10.04, maximum nuisance cosine 0.367, and augmented condition number 1.47. This is not an apparatus selection: magnetic pickup, trap transfer, vortex state, field-sham covariance, specimen impedance, and the state-conditioned Green response have not been measured.

For the hypothetical niobium coating, the condensation-energy screen $B_c^2 V/(2\mu_0 c^2)$ gives a mass equivalent 2.27×10^{-28} kg, or 7.33×10^{-13} of the probe mass. It is retained as an ordinary stress-energy upper bound, not inserted into the Diósi generator. The Anderson--Higgs/Standard-Model Higgs relation remains a structural analogy; BEC coherence remains a conditional replication-platform relation requiring a new many-body density contract. Neither supplies a collapse kernel.

5.9 Integrated empirical-pilot closure

The integrated readiness gate turns the remaining feasibility questions into one fail-closed same-apparatus packet contract. It preserves the leading design, public component replays, proper-time budget, and superconducting boundary control as immutable upstream evidence. Public measurements from different instruments remain useful component checks, but they are not fused into the proposed apparatus covariance or likelihood.

Two estimands are now made impossible to confuse. The primary Diósi test is a held-out contraction that follows the frozen mass--separation--hold-time law. For the leading apparatus,

$$V_{\mathrm{D}} = e^{-\Gamma_{\mathrm{D}} t} = e^{-0.0295115} = 0.970920, \qquad \sigma_{|C|} \leq \frac{1 - e^{-\Gamma_{\mathrm{D}} t}}{\mathrm{SNR}_{\min}} = 5.816 \times 10^{-3}. \tag{28}$$

The 2.908% contraction and precision target at the registered SNR floor of five are conditional predictions of the frozen effective-Gaussian model, not measured evidence.

The boundary-by-superposition estimator is instead the four-cell cross-ratio

$$R_4 = \frac{\bar{C}_{\mathrm{active,sep}} \bar{C}_{\mathrm{reference,compact}}}{\bar{C}_{\mathrm{active,compact}} \bar{C}_{\mathrm{reference,sep}}}. \tag{29}$$

A standard boundary-independent Diósi factor multiplies both separated cells and cancels exactly from R_4 when the active and reference cells share the same mass-density and branch-separation histories. Otherwise the two integrated Diósi exponents must be computed separately and cancellation is not asserted. Therefore the primary contraction tests the registered Diósi law, whereas non-unit R_4 tests boundary--superposition nonfactorization after the ordinary electromagnetic response is calibrated. Neither statistic alone establishes a Casimir-to-collapse mechanism.

The integrated feasibility pilot remains unauthorized because all eight joint authorities are absent: (1) state preparation/recombination plus the selected- specimen mass-density identity; (2) as-built geometry and finite Green response; (3) measured material spectra; (4) worldline and phase covariance; (5) the quantum gas-collision kernel; (6) low-statistics four-cell complex-coherence commissioning data; (7) an independently powered companion channel; and (8) an exact external-bound recast for the registered $R_0 = 100$ nm normalization.

Two numerical limits are design conventions, not physical constants or achieved tolerances. The 0.034644-rad phase ceiling was transported from the superseded parent forecast by allocating Gaussian phase-jitter loss to one tenth of that forecast's exponent,

$$\sigma_{\phi,\max} = \sqrt{2(0.1)(0.00600105)} = 0.034644 \text{ rad.}$$

It has not been reoptimized against the selected specimen or the 0.00435852 transported diagnostic exponent and must be refrozen in the pilot analysis plan. The 0.10 pilot-to-confirmatory relative covariance-drift ceiling is a provisional engineering convention; covariance uncertainty must be propagated and the threshold justified before confirmatory use. Blinding, train/holdout separation, custody, and zero cross-apparatus covariance fusion are mandatory.

5.10 Retarded-source propagation closes a missing ordinary-physics lane

The leading design does not treat the boundary label as a magical switch. Every time-dependent voltage, current, trap field, compensation channel, and readout field can propagate to the two branches with a delay, polarization, phase, and dissipative response. Those responses belong to the ordinary null before any remaining contraction is compared with the frozen Diósi law.

The familiar kinked-field-line construction is useful intuition, but field lines are not the dynamical derivation. In the nonrelativistic radiation limit, Maxwell's equations with a conserved source and retarded boundary condition give

$$\mathbf{E}_{\mathrm{rad}}(\mathbf{r},t) = \frac{q}{4\pi\epsilon_0 c^2 R}\,\hat{\mathbf{n}} \times \left[\hat{\mathbf{n}} \times \mathbf{a}(t-R/c)\right]. \tag{30}$$

The Stage-4.2S analytic benchmark recovers the transverse field, zero- acceleration limit, $1/R$ amplitude law, retarded delay, circular- polarization projector, current conservation, and Larmor power. Numerical angular integration agrees with the analytic radiated power to relative error 2.57×10^{-16}. This establishes software and equation recovery, not a measured radiation background.

The dimensionless propagation screen is

$$kL = \frac{2\pi f L}{c}. \tag{31}$$

For the frozen $f = 0.5$ Hz boundary label and $L = 80\,\mu\mathrm{m}$, $kL = 8.38 \times 10^{-13}$: geometric electromagnetic retardation at the fundamental is negligible. A synthetic 1550-nm optical benchmark gives $kL = 324.29$, so optical response requires a wave calculation. The registered 1-kHz switching-edge and 1-MHz RF rows are scale demonstrations only; they do not assert the final rise time, trap frequency, or readout wavelength. A slow fundamental also does not bound unmeasured switching harmonics, ringing, material relaxation, mechanical sidebands, or optical backaction.

The apparatus calculation must therefore use the measured source spectrum and the as-built retarded dyadic Green response:

$$E_i(\mathbf{r},\omega) = i\mu_0\omega \int d^3r'\, G^{\mathrm{ret}}_{ij}(\mathbf{r},\mathbf{r}',\omega) J_j(\mathbf{r}',\omega), \qquad C_{0,\beta} = C(0)e^{i\Phi_{\mathrm{EM},\beta} - \chi_{\mathrm{EM},\beta}}. \tag{32}$$

Stage 4.2S verifies, with a polarization-retaining synthetic Green matrix, the algebraic map from branch fields into differential energy and phase, absorption and heating, photon recoil, ordinary contraction, and the complex-coherence nuisance vector $(-\chi_{\mathrm{EM}}, \Phi_{\mathrm{EM}})$. Its numerical phase and loss are deliberately synthetic and are not apparatus forecasts.

Ordinary-null integration remains unauthorized because 0/7 authorities are ready: measured source current maps and waveforms; the as-built retarded Green tensor; measured complex material response; branch geometry and polarization transfer; switching-edge spectral coverage; joint phase/loss/recoil/heating covariance; and independent full-wave plus energy-balance verification. Consequently measured evidence and retarded-source covariance remain `not_ready`, residual attribution and collapse identification remain `blocked`, physical viability remains `not_evaluated`, and the physical pilot remains unauthorized. The frozen Diósi generator is unchanged and no radiation-, polarization-, Green-tensor-, frequency-, or Casimir-to-collapse edge is registered.

5.11 Why the selected power is still conditional

The response/covariance geometry used by the bounded search was transported from the same synthetic world used to establish the parent candidate family. Finite-pilot covariance uncertainty was not propagated. Accordingly, 0.927386 is a conditional calculation, not a robust power claim. The 0.997858 value is retained only as the parent-design result. The next empirical computation must report an envelope rather than a single number.

The prespecified stress family must include:

Stress axis	Required analysis	Current status
finite-pilot covariance	nested simulation, bootstrap, or analytical propagation	not run
shrinkage/regularization	repeat whitening across declared estimators	not run
drift and non-Gaussian tails	correlated drift and heavy-tail worlds	not run
response amplitude/phase error	perturb every nuisance vector within pilot uncertainty	not run
missing nuisance	inject unregistered channels and test residual inflation	not run
branch and hold-time jitter	propagate preparation/metrology distributions	not run
shared calibration ancestry	model cross-control correlation explicitly	partially represented, not stressed
leave-one-control-out	refit after withholding each control family	not run
covariance structure	alternate prespecified block families	not run
selection optimism	repeat candidate search in independent synthetic worlds	not run

The accepted future headline has the form: "Across the registered uncertainty family, power ranged from X to Y and all design gates survived in Z% of worlds." Until X, Y, and Z are computed, the acquisition-power claim remains provisional.

5.12 Confirmatory analysis plan still to be frozen

The primary confirmatory comparison will score the frozen M_0 and $M_0 + \mathrm{D}$ predictions in joint complex-coherence space with the pilot-frozen full covariance. Neither the Diósi amplitude nor R_0 may be fitted on confirmatory data. An amplitude-fitted curve may be shown only as a labeled diagnostic. The raw-complex likelihood is authoritative unless a Gaussian log-visibility likelihood demonstrates at least 95% coverage over the planned visibility range.

Nuisance responses, covariance regularization, exclusions, cell order, prediction vectors, and robustness thresholds must be learned from calibration and pilot partitions, then frozen before held-out acquisition. Confirmatory and independent-replication data may not refit them. A complete statistical analysis plan must additionally define the primary test statistic, calibrated decision threshold, interval construction, missing-shot and failed-preparation handling, stopping rule, multiplicity across R_4, companion, and diagnostic analyses, and the independent-replication criterion.

The quoted 1,028 paired allocation units and power 0.927 are planning values, not an acquisition plan. A paired unit is presently a dimensionless synthetic allocation unit; it has no frozen mapping to shots per cell, cycle duration, attrition, duty cycle, or wall-clock time. Those quantities and the complete m, d, t allocation must be fixed before the number can guide acquisition. Failure to recover the declared error guarantees under covariance uncertainty, non-Gaussian coverage, attrition, selection optimism, or a newly observed nuisance produces a statistically indeterminate result or apparatus no-go, not a measured null or positive collapse result.

6. External constraints and scientific reach

The selected $R_0 = 100$ nm point must be compared with constraints on the same model convention. Direct spontaneous-radiation searches currently provide a strong lower-bound axis for the nondissipative Markovian Diósi implementation. The 2026 XENONnT analysis reports $R_0 > 4.9 \times 10^{-10}$ m at 90% confidence and $R_0 > 4.5 \times 10^{-10}$ m at 95% confidence in its stated Diósi convention [7]. Earlier germanium searches ruled out the natural parameter-free proposal and progressively strengthened the lower limit [8,9].

The selected $R_0 = 1.0 \times 10^{-7}$ m is about 204 times the XENONnT 90%-confidence lower bound. It is therefore not excluded by that one-sided screening comparison. This is not yet a full admission receipt: the mass-density smearing definition, normalization convention, charged-constituent radiation map, and treatment of the effective composite particle must be matched line by line before the point is called externally allowed.

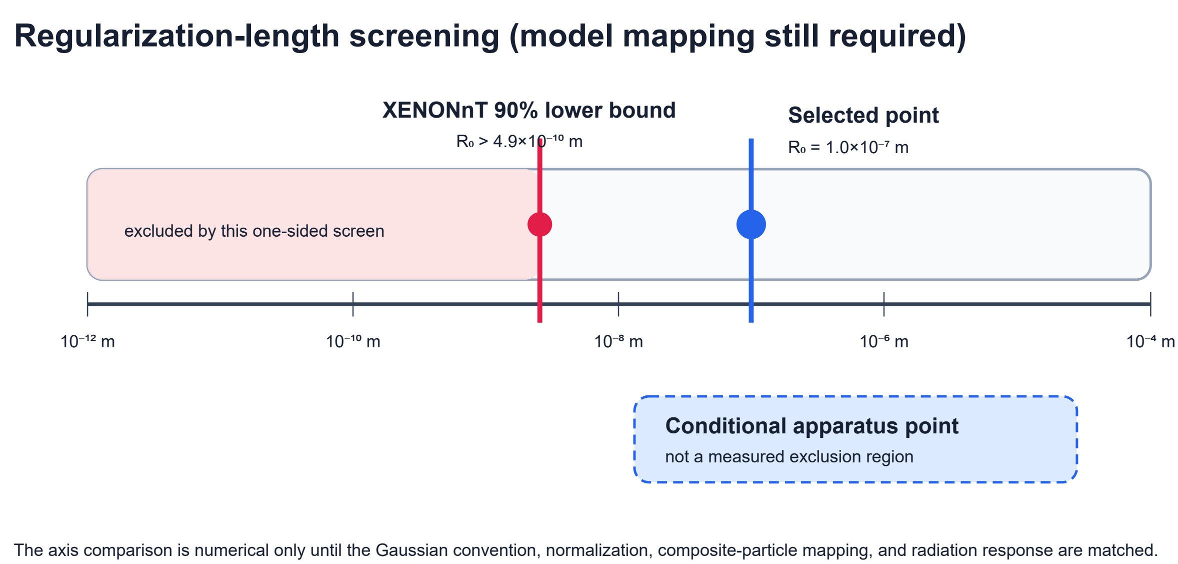

Figure 5. Logarithmic R0 screening. The XENONnT lower bound and the selected 100 nm point are shown on the same numerical axis. The shaded “apparatus” marker is conditional sensitivity, not a measured exclusion. The formal model-equivalence audit remains open.

The current reach statement is therefore limited:

- a measured null with validated sensitivity could exclude the registered effective-particle point;
- it would not exclude all regularized, dissipative, colored-noise, or Penrose-motivated models;
- a positive coherence residual could favor the frozen line shape over the registered nuisance span;

- it could not distinguish neighboring collapse generators without additional parameter and companion observables;
- a boundary-correlated excess would first challenge H0 and the joint-system equivalence assumption, not prove HB.

7. Leading-design forecast

Figure 6 shows the conditional coherence curve from Eq. (2). It contains no ordinary-decoherence band because that band must be learned from pilot data.

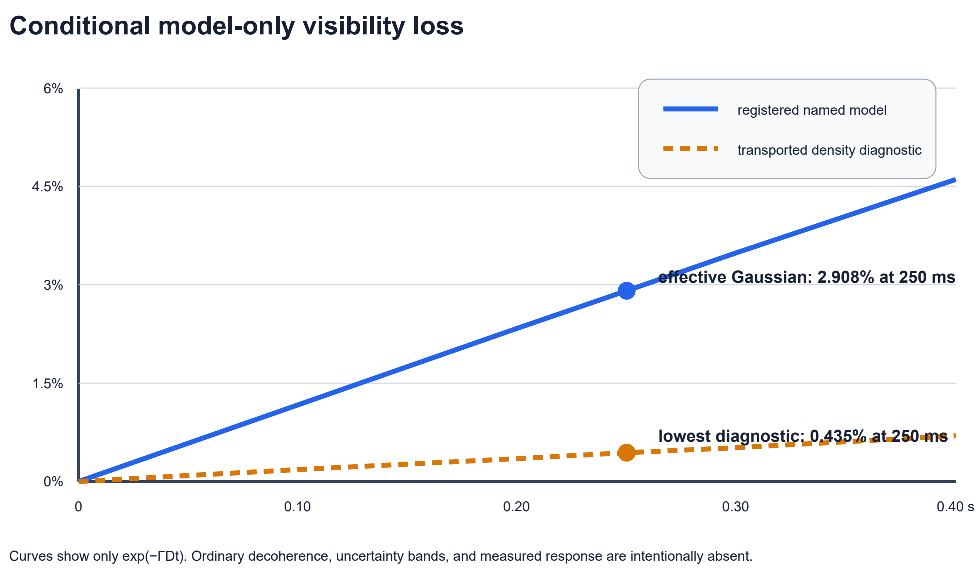

Figure 6. The registered effective-Gaussian curve reaches 2.908% loss at 250 ms, while the lowest transported density diagnostic evaluated so far reaches 0.435%. This is a model-sensitivity comparison, not a statistical confidence band or physical lower bound. Neither curve is a total visibility forecast.

7.1 Companion observable and interpretation ceiling

The same nondissipative generator predicts momentum diffusion [1--3,10]

$$D_{pp} = \frac{G\hbar m^2}{12\sqrt{\pi}R_0^3} \tag{33}$$

and a center-of-mass energy increase

$$\dot{E} = \frac{3D_{pp}}{m} = \frac{G\hbar m}{4\sqrt{\pi}R_0^3}. \tag{34}$$

For the leading manifest, $\dot{E} = 3.07013 \times 10^{-40}$ W. If 100 independent samples and SNR≥5 were demanded, the algebraic maximum one-shot standard uncertainty would be

$$\sigma_{\dot{E},\text{one shot}} \le \frac{\dot{E}\sqrt{100}}{5} = 6.14027 \times 10^{-40}\ \text{W}. \tag{35}$$

Equation (35) is not an instrument model. No registered detector demonstrates this bandwidth, calibration, independence, or noise floor. The companion observable is therefore presently a structural no-go for the paper's strongest positive interpretation.

This revision adopts the conservative third option:

> Unless an independently powered companion channel becomes feasible, a replicated Diósi-shaped coherence residual may be reported as unexplained model-consistent phenomenology, but not as support-eligible identification of the Diósi generator.

The project may replace the heating channel only through a prespecified derivation from the same frozen dynamics, with an instrument-level bandwidth, integration-time, noise, calibration-transfer, and cross-covariance model.

8. Commissioning, feasibility-pilot, and confirmatory decisions

8.1 Subsystem commissioning first

The presently authorized activity is subsystem commissioning, not a search for collapse and not the integrated feasibility pilot. It must deliver provenance-bound measurements of:

1. object mass, geometry, material, coating, and internal-density model;
2. state-preparation fidelity, an explicit identical-branch or sham-split control, and separated-branch metrology;
3. center-of-mass packet centers/covariances, overlap, momentum difference, hold-time jitter, and recombination metrology in both boundary states;
4. finite-geometry electromagnetic Green response and material permittivity;
5. boundary-switching transfer functions and settling time;
6. thermal, gas, vibration, tilt, patch, charge, optical, and sensor responses;
7. raw complex-coherence covariance, including shared calibration ancestry;
8. sham, detuned, echo, and path-swap controls;
9. companion-channel feasibility or the lower interpretation ceiling;
10. blinded data custody and independent solver replication.

The four-cell item at this stage is a low-statistics readout, randomization, and custody demonstration. It is not powered model comparison and cannot be reported as a collapse null or excess.

8.2 Integrated feasibility-pilot admission rule

$$\mathcal{G}_{\text{pilot}} = \mathcal{G}_{\text{preparation}} \wedge \mathcal{G}_{\text{response}} \wedge \mathcal{G}_{\text{covariance}} \wedge \mathcal{G}_{\text{identifiability}} \wedge \mathcal{G}_{\text{power}} \wedge \mathcal{G}_{\text{custody}}. \tag{36}$$

The integrated feasibility pilot is authorized only after all eight authority packets in Section 8.5 qualify. It is admitted to confirmatory planning only if every factor above is then true. The numerical identifiability gates remain Eq. (17), but power must additionally survive the uncertainty envelope in Section 5.10. If any factor fails, the result is an explicit apparatus-redesign no-go.

8.3 Confirmatory campaign

Only after pilot admission may the team freeze:

- one canonical apparatus identity;
- one mass-density representation and R_0 point or prespecified grid;
- exclusions and quality-control rules;
- response and covariance ancestry;
- analysis code and randomization;
- companion interpretation ceiling;
- primary and replication sample sizes.

The confirmatory data remain blinded until completeness and custody checks pass. An independent replication is required for any positive interpretation.

8.4 Decisive outcomes

Observed outcome	Defensible inference	Not established
pilot cannot prepare the state	current apparatus infeasible	collapse model false
pilot fails cosine/conditioning gates	current design non-identifiable	collapse model false
measured null with demonstrated sensitivity	registered point disfavored/excluded at stated confidence	all Diósi-family models or Penrose OR false
residual follows ordinary control	H0 response model repaired	collapse
corrected four-cell interaction is null	no resolved boundary--branch nonfactorization at demonstrated sensitivity	standard boundary-independent Diósi model false
corrected four-cell interaction is nonzero	H0 or complete-joint-system equivalence challenged	Casimir-modified collapse without a registered kernel
boundary-correlated residual without transfer kernel	anomaly eligible for a new bridge study	Casimir-induced collapse
replicated Diósi-shaped coherence residual, no companion	unexplained model-consistent phenomenology	generator identified
replicated shape plus independent companion	tested generator gains support within registered alternatives	manifold dynamics or universal objective collapse
powered null against a separately completed and frozen Penrose candidate	that candidate and powered parameter region excluded	every Penrose-inspired completion false
frozen Penrose-specific time shape plus same-kernel companion	that registered candidate gains support among tested alternatives	a unique quantum-gravity theory or direct observation of two manifolds

8.5 Expert collaboration work packages

We are not asking external readers to endorse the synthetic candidate. We are asking whether each same-apparatus authority packet below can be produced, and whether any packet falsifies the apparatus before an integrated feasibility pilot is attempted. All eight are presently absent.

Authority packet	Minimum contribution requested	Acceptance boundary
state preparation, recombination, and specimen identity	demonstrate branch separation, 250 ms hold, recombination, path swap, internal-state equivalence, fidelity, non-postselected failure accounting, measured object mass, and the specimen-specific mass-density/coating map used in E_G	measured on the leading state and selected specimen with a content-addressed receipt
as-built geometry and Green response	metrology-bound geometry including edges, apertures, branch vector, tilt, and nearby conductors; independently checked finite-geometry Green/scattering solution	computed from measured leading-apparatus geometry, with convergence and independent-solver checks
measured material spectral response	temperature-dependent complex dielectric or impedance data for the actual sphere, coatings, plates, and contamination state	same specimens and states used in acquisition, with calibration uncertainty
worldline and phase covariance	3D worldlines plus gap, tilt, vibration, voltage, patch, thermal, clock, readout, local-gravity, and echo-transfer covariance	measured on the leading apparatus; covariance ancestry frozen before held-out data
quantum gas-collision kernel	species-resolved pressure and temperature, scattering inputs, confinement geometry, momentum transfer, and independent pressure calibration	computed from measured gas conditions with uncertainty and zero-density limit
four-cell complex-coherence commissioning	low-statistics raw I/Q or equivalent data for active/reference × separated/compact cells, randomized acquisition, witnesses, exclusions, and held-out ancestry	all four cells from one apparatus with blinded labels and frozen covariance; readout/custody qualification only, not model scoring
independent companion channel	instrument-level same-generator observable with bandwidth, integration time, noise, calibration transfer, independence, and cross-covariance--or a documented physical no-go	measured/computed-from-measured receipt; no proxy SNR
exact external-bound recast	source-backed mapping of independent bounds to the exact nondissipative Gaussian Diósi convention and composite-particle representation at $R_0 = 100$ nm	published external evidence or independently audited recast

A qualifying packet is complete, content-addressed, and reviewed by an independent custodian. Worldline, four-cell, and companion packets must preserve frozen covariance ancestry. Except for the external-bound recast, every packet must refer to the leading apparatus. A partial packet set may motivate redesign but cannot authorize the integrated feasibility pilot; synthetic fixtures and measurements from other apparatuses do not substitute for a missing packet.

9. Observable-separation gate

Force, phase, coherence, heating, curvature, and collapse rate are different observables. They may be connected only by a declared model with units, parameters, provenance, and falsifiers. The governing gate is:

$$\text{observable A} \to \text{observable B} \quad \Longrightarrow \quad \{\text{transfer law, units, source, parameters, test}\}. \tag{37}$$

Equation similarity, shared use of Planck's constant, or a common energy-frequency conversion is insufficient. In particular:

- Casimir pressure is not Diósi self-energy;

- renormalized negative energy density is not automatically negative spacetime curvature for the complete apparatus;
- virtual-particle language is not a detector model;
- gravitational-wave interference does not imply phase coherence between Casimir and matter-wave channels;
- circular polarization is an electromagnetic control degree of freedom, not a collapse source;
- a light cone defines causal structure, not a nonrelativistic collapse generator.

10. Limitations

The main limitations are scientific, not typographic:

1. **State preparation:** the leading mass is approximately 1.10 million times the 170 kDa free-particle interference benchmark; measured packet width, overlap, trajectory, and boundary-to-boundary branch equivalence are absent.
2. **Synthetic optimization:** the selected design was evaluated in the same assumed response/covariance world in which it was found.
3. **External-bound mapping:** the selected R0 passes a numerical lower-bound screen, but the exact convention/composite mapping is not certified.
4. **Mass representation:** the 2.908% prediction is specific to one effective Gaussian particle; the transported representation envelope lowers the transported diagnostic benchmark to 0.435%, while selected-specimen homogeneous, layered, coarse-grained, and atomistic calculations are incomplete.
5. **Companion channel:** the heating signal lacks a physically credible detector receipt, lowering the maximum positive interpretation.
6. **Ordinary physics:** finite-geometry material and environmental responses are unmeasured for the integrated apparatus.
7. **Relativity:** the registered dynamics is nonrelativistic and does not compute manifold evolution.
8. **Boundary bridge:** no Casimir-to-collapse transfer kernel exists in the tested model.

The machine-readable status is equivalent to the following prose: measured evidence is not yet available; empirical pilot readiness and physical viability have not been established; collapse identification and manifold dynamics remain blocked.

11. Discussion

11.1 What is new

The useful result is not that a sufficiently massive object has a large collapse rate; that scaling is already built into the selected model. The advance is the conversion of that rate into a fail-closed experimental comparison. A candidate signal is expressed in the same complex-coherence coordinates as the measured control responses, whitened by their joint covariance, and tested for linear dependence before acquisition power is reported. The rejected design demonstrates why this order matters. A nominally nonzero loss can be statistically powerless as evidence when its direction is indistinguishable from ordinary drift or decoherence.

The individual ingredients are not claimed as novel: matter-wave collapse-model reach, nanoparticle preparation sequences, Casimir-systematic accounting, response-function metrology, and staged pathfinder programs all have established precedents [24--29]. The contribution is their synthesis into two explicitly different estimands--a model-specific m, d, t contraction and a boundary nonfactorization ratio--with complex-data retention, covariance-aware identifiability, packet-level authority, and an inference rule that leaves an unexplained residual unexplained.

This approach also changes what counts as a useful negative result. Failure to prepare the state is an apparatus no-go. Failure of the cosine or conditioning gate is an identifiability no-go. A measured null becomes a model constraint only after the apparatus demonstrates sensitivity to the frozen direction. These outcomes answer different questions and should not be combined into a single “experiment failed” category.

11.2 Why retain the Casimir boundary

The Casimir boundary remains valuable even though it does not enter Eq. (2). It creates a tunable electromagnetic and material environment with strong gap, temperature, polarization, and surface dependence. That makes it an unusually demanding test of whether the ordinary-physics null is complete. Sham switching, detuning, static gap cells, path swaps, and polarization controls can reveal phase rotations or losses that might otherwise be misidentified as a mass-dependent collapse residual.

This use is scientifically asymmetric. Agreement with finite-geometry macroscopic QED improves confidence in H0; disagreement produces a boundary-response anomaly. Neither result modifies HD automatically. Only a new, independently motivated transfer law could make the boundary a parameter of a collapse generator. Keeping that bridge closed protects both sides of the experiment: ordinary boundary physics is not mislabeled as gravity, and a failure of the particular Diósi model is not mislabeled as a failure of Casimir theory.

11.3 What a positive result would mean

A coherence residual with the registered mass, separation, and hold-time dependence would be more informative than an unexplained scalar decay rate. Replication across object masses and branch geometries would make an ordinary single-channel explanation progressively less plausible. Even so, the interpretation remains conditional on the alternative set. An unmodeled gas, patch, readout, preparation, or calibration response can imitate a structured signal if it shares the same experimental schedule.

The companion channel was intended to reduce that ambiguity by demanding a second prediction from the same generator. Its forecast, however, is too small to count as an experimental discriminator without a credible instrument proposal. The present claim ceiling is therefore deliberately lower than the original protocol's aspirational ceiling. A replicated coherence residual without a companion would motivate new tests and model comparison; it would not identify objective collapse. This is a scientific consequence of the detector gap, not merely cautious wording.

11.4 What a null result would mean

At a demonstrated sensitivity, a null result can exclude the registered effective-particle point. It cannot directly exclude a rigid-sphere representation that was never converged, a different regularization length, a dissipative or colored-noise theory, or Penrose's broader lifetime motivation. The mass-density robustness campaign therefore determines how widely the null may be interpreted. If several defensible representations predict comparable losses and all are within reach, the experiment tests a family. If their predictions diverge substantially, the representation itself becomes part of the prespecified alternative set.

11.5 Priority order after this design study

The next scientific work should be ordered by the chance that it invalidates the apparatus before costly acquisition:

1. demonstrate or reject state preparation at the registered mass, separation, and hold time;
2. replace the conservative residual-gas no-go with a measured composition, pressure, and scattering/collision-veto model or redesign the apparatus;
3. complete provenance-bound layered and constituent density maps;
4. complete the exact external-bound convention and composite-particle map;
5. measure finite-geometry material and boundary transfer functions;
6. obtain pilot response vectors and covariance;
7. run the registered robustness envelope and repeat candidate selection in independent synthetic worlds;
8. either demonstrate a companion instrument or retain the lower interpretation ceiling;
9. only then freeze a blinded confirmatory campaign.

This sequence keeps the paper's methodological core useful even if the present apparatus is rejected. The estimator, hypothesis separation, packet custody, and design gates can be carried into a lower-mass or otherwise redesigned platform without preserving the current sphere as a favored physical object.

11.6 Supporting derivations and design ancestry

The full supporting derivations and historical apparatus narrative are preserved in Sections B.13--B.15 of the reproducibility supplement. The compact record below retains the equations needed to understand why the present design was selected.

Electromagnetic and environmental closure. For an isotropic ground state, the ordinary macroscopic-QED chain is [30]

$$\alpha_g(i\xi) = \frac{2}{3\hbar}\sum_n \frac{\omega_{ng}|\langle n|\hat{\mathbf{d}}|g\rangle|^2}{\omega_{ng}^2+\xi^2}, \qquad U_{\mathrm{CP}}(\mathbf{r}) = \frac{\hbar\mu_0}{2\pi}\int_0^\infty d\xi\,\xi^2\alpha_g(i\xi)\,\mathrm{Tr}\,\mathbf{G}^{(1)}(\mathbf{r},\mathbf{r};i\xi).$$

For the diagnostic small-sphere limit,

$$\alpha_{\mathrm{sph}}(i\xi) = 4\pi\epsilon_0 R^3\frac{\epsilon(i\xi)-1}{\epsilon(i\xi)+2}.$$

Charge/current response controls this Casimir lane; mass density controls the registered Diósi lane. Phase fluctuations and gas collisions enter separately:

$$\left\langle e^{i\delta\phi}\right\rangle = e^{-\sigma_\phi^2/2}, \qquad \chi_\phi = \frac{\sigma_\phi^2}{2}.$$

$$\Gamma_{\mathrm{gas}}(\Delta\mathbf{x}) = n_g\int d^3v\,\mu(\mathbf{v})v\int d\Omega\,|f(\mathbf{q},\mathbf{v})|^2\left[1-e^{i\mathbf{q}\cdot\Delta\mathbf{x}/\hbar}\right].$$

The superseded silica calculation established two design consequences: a normal split can generate an overwhelming boundary phase, whereas ideal-plane tangential symmetry cancels the nominal differential phase but not sensitivity to alignment noise. The selected specimen still requires a finite-geometry Maxwell solution, measured spectra, and a species-resolved gas kernel.

Geometry and representation audit. The finite-rectangle surrogate and its phase-covariance map are

$$\Omega(\mathbf{r}) = \iint_A \frac{z\,dA'}{[(x'-x)^2+(y'-y)^2+z^2]^{3/2}}, \qquad U_{\mathrm{rect}}(\mathbf{r}) = \frac{\Omega(\mathbf{r})}{2\pi} U_{\mathrm{CP}}^{\infty}(z).$$

$$J_i = \frac{\partial \Phi_{\mathrm{EM}}}{\partial \theta_i}, \qquad \sigma_\phi^2 = \mathbf{J}\,\mathbf{\Sigma}_\theta \mathbf{J}^\top.$$

The isotropic gas proxy and density-representation screen are

$$\Gamma_{\mathrm{gas}}(d) = \sum_s n_s \bar{v}_s \sigma_s \frac{1}{2} \int_{-1}^{1} d\mu \left[1 - \mathrm{sinc}\left(\frac{m_s \bar{v}_s d\sqrt{2(1-\mu)}}{\hbar}\right)\right].$$

$$E_G^{(a)}(d, r_0) = \frac{2Gm^2}{\pi r_0} \int_0^\infty du\, e^{-u^2} |F_a(uR/r_0)|^2 \left[1 - \mathrm{sinc}(ud/r_0)\right].$$

These diagnostics rejected the parent tolerances, showed a factor-6.77 spread across the registered density profiles, and exposed the preparation-scale gap. They do not select the physical density profile or establish achieved environmental control.

Bounded apparatus selection. The search uses the conjunction

$$G_M(q) = G_{\mathrm{D}} \wedge G_\rho \wedge G_\phi \wedge G_{\mathrm{gas}} \wedge G_{\mathrm{prep}} \wedge G_{\mathrm{bound}} \wedge G_{\mathrm{id}} \wedge G_{\mathrm{power}} \wedge G_{\mathrm{companion}},$$

with transported backgrounds

$$\sigma_{\phi,\mathrm{echo}}^2 = \eta_{\mathrm{echo}}^2 \mathbf{J}_\phi(q) \mathbf{\Sigma}_q \mathbf{J}_\phi^\top(q), \qquad \Gamma_{\mathrm{gas}}(q) = \Gamma_{\mathrm{gas,L}}\, \mathcal{T}_{P,T,R,d}(q).$$

Three of 200 candidates survived the bounded synthetic filters. The best-ranked diamond-density point is the commissioning target summarized in Sections 1.2 and 5.4. The apparent companion value is only a scaling surrogate; measured response, covariance, gas, preparation, companion-detector, and exact-bound authority remain absent. Thus the search selects what to commission, not an experiment-ready apparatus or a physical result.

12. Claim boundaries

This paper establishes:

- the exact model and parameter point used by the forecast;
- the failure of the first synthetic identifiability design;
- the nominal separability of a bounded redesign under stated assumptions;
- a leading synthetic design with a 2.908% effective-Gaussian prediction and a 0.435% lowest transported density diagnostic evaluated so far;

- a selected-candidate forecast of 1,028 dimensionless paired allocation units and power 0.927 at the 1,600-unit ceiling;
- a representation envelope showing that the effective-Gaussian prediction is not representation-independent;
- a transported gas screen that passes at the leading 10^{-15} Pa target while remaining unmeasured;
- an ideal-plane orientation screen showing that boundary phase and phase jitter can dominate the target residual unless geometry is frozen and empirically calibrated;
- a frozen 3D search geometry, a crosschecked finite-plate surrogate, and explicit phase-control tolerances that define the selected echo/alignment target;
- a QLBE-structured gas proxy, current scalar external-bound screen, and four-representation Diósi sensitivity envelope;
- a bounded 200-candidate redesign search with three points that survive the synthetic ranking filters, while physical companion authority remains absent;
- a synthetic normal/superconducting control assessment that preserves finite- frequency impedance, rejects thermally collinear and fabrication-degenerate toggles, and verifies cancellation of standard boundary-independent Diósi attenuation from the boundary ratio;
- an executable integrated-pilot contract that separates the primary Diósi contraction from the four-cell boundary interaction and records a fail-closed readiness no-go while all eight same-apparatus authorities are absent;
- a fail-closed pilot and confirmatory decision structure;
- a Stage-0.1 synthetic relational-correspondence benchmark that preserves the known branch displacement, survives the frozen local invariance tests, and recovers the registered Gaussian Newtonian target to a relative discrepancy of 1.88×10^{-7}, while remaining blocked with 0 of 5 physical reference packets ready;
- the empirical and computational work still required.

This paper does not establish:

- that the apparatus can be built or the state prepared;
- that the nominal power survives model uncertainty;
- that the selected point is fully admitted by all external constraints;
- that a Casimir boundary changes a collapse rate;
- that a superconducting condensate, the Standard-Model Higgs field, or a BEC order parameter changes the registered Diósi rate;
- that the synthetic magnetic-toggle candidate is apparatus-ready;
- that objective collapse occurs;
- that Penrose's geometric interpretation is observed;
- that the Stage-0.1 local affine prescription is a unique or measured physical correspondence, a covariant incompatibility functional, or a reduction law;
- that spacetime manifold dynamics have been measured or simulated.

13. Conclusion

The strongest result is methodological. A scalar loss rate that is nearly collinear with ordinary backgrounds is not an adequate collapse test. The experiment must operate in raw complex-coherence space, measure the nuisance responses and covariance, project the frozen candidate away from that span, and fail closed when the geometry is ill-conditioned.

The bounded search yielded three synthetic survivors; the current commissioning manifest is its best-ranked point, not the earlier silica reference. It is a diamond-density 276.302 nm-radius sphere, 250 nm tangential split, 250 ms hold, 10 μm gap, 4 K, and 10^{-15} Pa. The frozen effective-Gaussian model predicts 2.908% loss; the lowest transported mass-density diagnostic evaluated so far gives 0.435%. In the registered whitened space, the point retains cosine 0.7177 and condition number 6.53. The synthetic calculation assigns 1,028 dimensionless paired allocation units and gives conditional forecast power 0.927 at 1,600 units.

The earlier no-go calculations are the evidence that shaped this design. They showed that the first signature was non-identifiable, the silica parent was gas dominated at its declared pressure, normal boundary orientation created an overwhelming phase screen, centered tangential cancellation was tolerance sensitive, and the Diósi energy varied by a factor of 6.77 across registered mass profiles. Those results are retained as derivation ancestry rather than presented as competing current apparatuses.

The leading design is still not physically viable evidence. Its electromagnetic response, covariance, gas kernel, echo rejection, measured worldlines and gravity covariance, state preparation, material density, external-bound mapping, and companion detector are transported or unmeasured. It therefore authorizes measured subsystem commissioning only, not an integrated feasibility pilot, confirmatory campaign, collapse claim, or manifold claim.

The Penrose candidate preflight and its new Stage-0.1 benchmark sharpen the theoretical destination without changing that standing. Stage-0.1 shows that one branch-blind local prescription survives the registered synthetic correspondence tests and recovers the existing Newtonian target. Because its five physical reference packets are absent, it does not clear the parent scientific correspondence blocker. Until that and the subsequent invariant- functional, causal-dynamics, probability, conservation, recovery, coherence, and companion gates close, the experiment supplies constraints that a missing theory must satisfy rather than evidence that the missing theory has been found.

The maintained runtime receipts record passing software, contract, equation- sidecar, theory-graph, and adapter checks for the design ancestry. Detailed test counts, run identifiers, hashes, and certificate records are kept in the reproducibility supplement and machine-readable receipts. They establish repository convergence only; they supply none of the eight missing empirical packets and do not validate a physical collapse, boundary bridge, or manifold claim.

The appropriate next action is therefore to complete the eight authority packets in Section 8.5. If the subsequently authorized integrated feasibility pilot reproduces the response geometry and survives the registered robustness family, a blinded confirmatory campaign becomes justified. If it does not, the apparatus is redesigned or rejected without converting an engineering failure into a claim about nature.

Publication declarations

Author contributions. To be completed by the human authors before submission. Repository history records the provenance of calculations and AI-assisted drafting; it does not determine scholarly authorship.

Code and data availability. The executable configurations, synthetic fixtures, reports, tests, equation-action sidecars, and receipts are maintained in the CasimirBot repository. The complete development record is preserved in the reproducibility supplement. No measured experimental dataset is reported. Before journal submission or public confirmatory registration, this statement must be completed with a public repository URL, archived DOI, release or commit identifier, software license, and reproducible environment manifest.

Competing interests. To be declared by the human authors before submission.

Acknowledgments. To be completed before submission.

References

1. L. Diósi, "A universal master equation for the gravitational violation of quantum mechanics," *Physics Letters A* **120**, 377–381 (1987). https://doi.org/10.1016/0375-9601(87)90681-5
2. L. Diósi, "Models for universal reduction of macroscopic quantum fluctuations," *Physical Review A* **40**, 1165–1174 (1989). https://doi.org/10.1103/PhysRevA.40.1165
3. A. Bassi, K. Lochan, S. Satin, T. P. Singh, and H. Ulbricht, "Models of wave-function collapse, underlying theories, and experimental tests," *Reviews of Modern Physics* **85**, 471–527 (2013). https://doi.org/10.1103/RevModPhys.85.471
4. R. Penrose, "On gravity's role in quantum state reduction," *General Relativity and Gravitation* **28**, 581–600 (1996). https://doi.org/10.1007/BF02105068
5. R. Penrose, *The Road to Reality* (Jonathan Cape, London, 2004), Chapter 30.
6. S. Pedalino *et al.*, "Probing quantum mechanics with nanoparticle matter-wave interferometry," *Nature* **649**, 866–870 (2026). https://doi.org/10.1038/s41586-025-09917-9
7. E. Aprile *et al.* (XENON Collaboration), "Challenging Spontaneous Quantum Collapse with the XENONnT Dark Matter Detector," *Physical Review Letters* **136**, 120201 (2026). https://doi.org/10.1103/2jm3-4976
8. S. Donadi *et al.*, "Underground test of gravity-related wave function collapse," *Nature Physics* **17**, 74–78 (2021). https://doi.org/10.1038/s41567-020-1008-4
9. I. J. Arnquist *et al.* (Majorana Collaboration), "Search for Spontaneous Radiation from Wave Function Collapse," *Physical Review Letters* **129**, 080401 (2022); Erratum **130**, 239902 (2023). https://doi.org/10.1103/PhysRevLett.129.080401
10. M. Carlesso, S. Donadi, L. Ferialdi, M. Paternostro, H. Ulbricht, and A. Bassi, "Present status and future challenges of non-interferometric tests of collapse models," *Nature Physics* **18**, 243–250 (2022). https://doi.org/10.1038/s41567-021-01489-5
11. H. B. G. Casimir, "On the attraction between two perfectly conducting plates," *Proceedings of the Royal Netherlands Academy of Arts and Sciences* **51**, 793–795 (1948).
12. E. M. Lifshitz, "The theory of molecular attractive forces between solids," *Soviet Physics JETP* **2**, 73–83 (1956).

13. M. Arndt and K. Hornberger, “Testing the limits of quantum mechanical superpositions,” *Nature Physics* **10**, 271–277 (2014). https://doi.org/10.1038/nphys2863
14. R. Colella, A. W. Overhauser, and S. A. Werner, “Observation of Gravitationally Induced Quantum Interference,” *Physical Review Letters* **34**, 1472–1474 (1975). https://doi.org/10.1103/PhysRevLett.34.1472
15. A. Roura, “Gravitational redshift in quantum-clock interferometry,” *Physical Review X* **10**, 021014 (2020). https://doi.org/10.1103/PhysRevX.10.021014
16. M. Zych, F. Costa, I. Pikovski, and Č. Brukner, “Quantum interferometric visibility as a witness of general relativistic proper time,” *Nature Communications* **2**, 505 (2011). https://doi.org/10.1038/ncomms1498
17. I. Pikovski, M. Zych, F. Costa, and Č. Brukner, “Universal decoherence due to gravitational time dilation,” *Nature Physics* **11**, 668–672 (2015). https://doi.org/10.1038/nphys3366
18. T. Bothwell *et al.*, “Resolving the gravitational redshift across a millimetre-scale atomic sample,” *Nature* **602**, 420–424 (2022). https://doi.org/10.1038/s41586-021-04349-7
19. P. W. Anderson, “Plasmons, Gauge Invariance, and Mass,” *Physical Review* **130**, 439–442 (1963). https://doi.org/10.1103/PhysRev.130.439
20. D. C. Mattis and J. Bardeen, “Theory of the Anomalous Skin Effect in Normal and Superconducting Metals,” *Physical Review* **111**, 412–417 (1958). https://doi.org/10.1103/PhysRev.111.412
21. M. J. A. de Jong *et al.*, “Measurement of a strong nonlinear force between superconductors compatible with the Casimir force,” *Nature Communications* **17** (2026). https://doi.org/10.1038/s41467-026-75261-9
22. D. M. Harber *et al.*, “Measurement of the Casimir–Polder force through center-of-mass oscillations of a Bose–Einstein condensate,” *Physical Review A* **72**, 033610 (2005). https://doi.org/10.1103/PhysRevA.72.033610
23. P. W. Higgs, “Broken Symmetries and the Masses of Gauge Bosons,” *Physical Review Letters* **13**, 508–509 (1964). https://doi.org/10.1103/PhysRevLett.13.508
24. R. Kaltenbaek *et al.*, “Macroscopic Quantum Resonators (MAQRO): 2015 update,” *EPJ Quantum Technology* **3**, 5 (2016). https://doi.org/10.1140/epjqt/s40507-016-0043-7
25. J. Bateman, S. Nimmrichter, K. Hornberger, and H. Ulbricht, “Near-field interferometry of a free-falling nanoparticle from a point-like source,” *Nature Communications* **5**, 4788 (2014). https://doi.org/10.1038/ncomms5788
26. G. Gasbarri, A. Belenchia, M. Paternostro, and H. Ulbricht, “Prospects for near-field interferometric tests of collapse models,” *Physical Review A* **103**, 022214 (2021). https://doi.org/10.1103/PhysRevA.103.022214
27. R. I. P. Sedmik and M. Pitschmann, “Next Generation Design and Prospects for CANNEX,” *Universe* **7**, 234 (2021). https://doi.org/10.3390/universe7070234
28. M. Abe *et al.*, “Matter-wave Atomic Gradiometer Interferometric Sensor (MAGIS-100),” *Quantum Science and Technology* **6**, 044003 (2021). https://doi.org/10.1088/2058-9565/abf719
29. E. Calloni *et al.*, “The Archimedes experiment,” *Nuclear Instruments and Methods in Physics Research A* **824**, 646–647 (2016). https://doi.org/10.1016/j.nima.2015.09.071
30. S. Scheel and S. Y. Buhmann, “Macroscopic quantum electrodynamics--concepts and applications,” *Acta Physica Slovaca* **58**, 675–809 (2008). https://arxiv.org/abs/0902.3586

31. S. Pedalino *et al.*, “Probing quantum mechanics with nanoparticle matter-wave interferometry - Dataset and code,” Zenodo data record (2025). https://doi.org/10.5281/zenodo.17502163
32. M. J. A. de Jong *et al.*, “Data supporting the paper ‘Measurement of the Casimir force between superconductors’,” Zenodo data record (2026). https://doi.org/10.5281/zenodo.18682702
33. NASA High Energy Astrophysics Science Archive Research Center, “LPFFILES--LISA Pathfinder Archive Data Summary.” https://heasarc.gsfc.nasa.gov/w3browse/all/lpffiles.html
34. R. Penrose, “On the Gravitization of Quantum Mechanics 1: Quantum State Reduction,” *Foundations of Physics* **44**, 557--575 (2014). https://doi.org/10.1007/s10701-013-9770-0
35. F. Giacomini and Č. Brukner, “Quantum superposition of spacetimes obeys Einstein's equivalence principle,” *AVS Quantum Science* **4**, 015601 (2022). https://doi.org/10.1116/5.0070018
36. W. C. Myrvold, “Relativistically invariant Markovian dynamical collapse theories must employ nonstandard degrees of freedom,” *Physical Review A* **96**, 062116 (2017). https://doi.org/10.1103/PhysRevA.96.062116
37. B. A. Juárez-Aubry, B. S. Kay, and D. Sudarsky, “Generally covariant dynamical reduction models and the Hadamard condition,” *Physical Review D* **97**, 025010 (2018). https://doi.org/10.1103/PhysRevD.97.025010
38. B. L. Hu and E. Verdaguer, “Stochastic Gravity: Theory and Applications,” *Living Reviews in Relativity* **11**, 3 (2008). https://doi.org/10.12942/lrr-2008-3
39. B. Dittrich, "Partial and Complete Observables for Canonical General Relativity," *Classical and Quantum Gravity* **23**, 6155--6184 (2006). https://doi.org/10.1088/0264-9381/23/22/006
40. J. Tambornino, "Relational Observables in Gravity: a Review," *SIGMA* **8**, 017 (2012). https://doi.org/10.3842/SIGMA.2012.017
41. J. D. Brown and K. V. Kuchar, "Dust as a Standard of Space and Time in Canonical Quantum Gravity," *Physical Review D* **51**, 5600--5629 (1995). https://doi.org/10.1103/PhysRevD.51.5600
42. K. Giesel and T. Thiemann, "Scalar Material Reference Systems and Loop Quantum Gravity," *Classical and Quantum Gravity* **32**, 135015 (2015). https://doi.org/10.1088/0264-9381/32/13/135015